

<https://doi.org/10.1038/s41746-025-02072-5>

Interpretable Multiomics Models for Predicting Surgical Interventions and Blood Transfusion Requirements in Traumatic Brain Injury

Check for updates

Jiang Deng^{1,8}✉, Tao Deng^{2,8}, Yan-Chun Zhang^{3,8}, Ning Zhao^{1,8}, Kai Liu^{4,5}, Chao-Jie Wang^{1,6}, Zi-Jian Zhang¹, Jing-Han Ma^{4,5}, Hao Wang^{4,7}, Li-Ping Lv¹, Ping Ma¹, Xiang-Yan Huang^{4,7}✉, Tao Wu³✉ & Yan-Yu Zhang¹✉

Accurately predicting surgical and transfusion needs in traumatic brain injury (TBI) patients remains challenging in emergency settings. We developed multiomics data fusion (MDF) models integrating clinical biomarkers, neural radiological imaging, and clinical text mining for predicting surgical intervention and blood transfusion requirements across four multicenter cohorts ($N = 2219$). The MDF models provided predictions a median of 3 hours before interventions, with surgical model F1 scores of 0.63–0.85 across external testing, outperforming single-domain approaches. The transfusion model demonstrated strong cross-center performance (validation/external F1 scores: 0.78, 0.74) and correlated well with actual transfusion volumes (validation: $R = 0.687$, external: $R = 0.580$). SHapley additive exPlanations revealed radiological features drove surgical predictions, while clinical parameters (lactate, GCS scores, pupillary reflex, and hemoglobin) were crucial for transfusion predictions. We also developed a simplified emergency model maintaining robust performance (validation AUC: 0.81, external AUC: 0.75). These models demonstrate cross-center generalizability and practical utility for emergency settings, supporting clinical implementation for improved TBI patient management.

Traumatic brain injury (TBI) is a leading cause of death and disability worldwide, particularly during major disasters, military conflicts, and emergency care settings^{1–4}. It is estimated that there are 69 million new cases globally each year², with mortality rates as high as 16% in elderly patients⁵. The pathophysiology of TBI involves direct mechanical injuries and secondary brain damage, initiating cascading pathophysiological changes that threaten patient survival and long-term quality of life^{6–8}. In emergency care, rapid identification of patients requiring surgical intervention and prediction of perioperative massive transfusion requirements are critical for optimizing treatment decisions and improving patient outcomes^{9–11}. However, timely decisions regarding these critical clinical interventions remain challenging in resource-limited emergency settings. Recent large-

scale randomized controlled trials have demonstrated that even under strict restrictive transfusion strategies, 38.4% (141/367)¹² and 48.5% (205/423)¹³ of TBI patients still required blood transfusion, highlighting the clinical importance of transfusion requirement prediction. Early identification of patients at high risk for massive transfusion within 2–4 h of admission is critically important for proactive blood bank preparation, timely blood resource allocation, and ensuring patient safety—particularly crucial in emergency settings where blood supply constraints and rapid decision-making demands intersect¹⁴. This early prediction enables medical teams to secure adequate blood reserves before surgical intervention, potentially reducing perioperative mortality and improving patient outcomes through timely transfusion when clinically indicated. Therefore, developing

¹Academy of Military Medical Sciences, Beijing, China. ²Department of Anesthesiology, The Second Hospital of Hebei Medical University, Shijiazhuang, China. ³Department of Blood Transfusion Medicine, The Seventh Medical Center of PLA General Hospital, Beijing, China. ⁴The 960th Hospital of the PLA Joint Logistics Support Force, Jinan, China. ⁵Department of Radiology, The 960th Hospital of the PLA Joint Logistics Support Force, Jinan, China. ⁶College of Biotechnology, Tianjin University of Science & Technology, Tianjin, China. ⁷Department of Blood Transfusion Medicine, The 960th Hospital of the PLA Joint Logistics Support Force, Jinan, China. ⁸These authors contributed equally: Jiang Deng, Tao Deng, Yan-Chun Zhang, Ning Zhao. ✉e-mail: ammsdjm@163.com; xiangyan73@aliyun.com; wut7175@sina.com; swgczhyy@126.com

predictive models capable of early identification of surgical indicators and prediction of massive transfusion risk is of significant importance for improving patient outcomes through ensuring timely and appropriate clinical interventions and optimal blood resource utilization.

Recent advancements in machine learning (ML), deep learning, and natural language processing (NLP) have demonstrated the immense potential of these tools in the medical field, particularly in the development of automated clinical decision support models^{15–18}. Traditional approaches relying on clinical biomarker analysis are increasingly being supplemented and enhanced by radiomics and deep learning techniques^{19,20}, with particularly active applications in the TBI field^{21–24}. Additionally, NLP-based deep learning models can extract key information from electronic health records, providing valuable solutions for disease prediction and treatment planning^{25–27}. However, despite these technological advances, research on their applications in TBI management—particularly in automated prediction of surgical indications and perioperative transfusion requirements—remains relatively limited. While some predictive models for surgical indications^{28,29} and transfusion¹⁶ based on traditional clinical indicators have shown good performance, these studies are mostly limited to single-center validation and lack extensive multi-center validation in heterogeneous patient populations. More importantly, according to the Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD) + AI statement guidelines³⁰, these models must undergo rigorous evaluation across multiple dimensions, including discrimination, calibration, and clinical utility—critical metrics that are frequently underreported in existing literature^{30–34}.

The integration of multicenter, multimodal data has emerged as a powerful approach in biomedical research, offering significant benefits in improving prediction model robustness and cross-scenario applicability^{35–37}. While concerns about model complexity versus clinical practicality are valid, existing single-modality prediction models for TBI management demonstrate significant limitations in real-world applications, particularly regarding cross-institutional generalizability and performance consistency across diverse patient populations. In the context of TBI management, multimodal models that synthesize clinical phenotypic data, radiological imaging findings, and medical text data provide a more comprehensive understanding of the complex pathophysiology than single-domain prediction models do, theoretically enabling more precise clinical decision-making. However, the increased sophistication of these models introduces new challenges, not only regarding model interpretability and clinical trust, but also including increased computational complexity from multimodal data fusion, potential information redundancy issues, and practical deployment difficulties in clinical environments. To address these challenges, on one hand, interpretable machine learning techniques have become crucial tools for enhancing model transparency by revealing decision-making processes and quantifying the contribution of individual input variables to prediction outcomes, thereby improving clinical trust^{38,39}; on the other hand, it is necessary to reduce the technical barriers for practical deployment through optimizing model architecture and reducing feature dimensionality. Consequently, the development of predictive models that balance prediction performance, deployment feasibility, clinical interpretability, and multicenter applicability is essential for advancing precision TBI management.

To address these challenges in TBI management, this study integrates early-available clinical biomarkers, radiomic features, and clinical text data to develop multimodal prediction models with automated decision support capabilities for surgical intervention indications and massive blood transfusion requirements (defined as ≥ 4 U RBCs based on previous studies^{40–42}) during the early admission period (within 2–4 h). We constructed a multimodal dataset comprising four distinct cohorts with 2219 patients and validated the predictive performance of our models on this comprehensive dataset (Table 1). To enhance clinical credibility and transparency, we incorporated interpretable machine learning techniques, including SHapley Additive exPlanations (SHAP) and Gradient-weighted Class Activation Mapping (Grad-CAM), to elucidate model prediction mechanisms and

provide individualized decision explanations. Additionally, considering the diverse needs of different clinical scenarios, we developed simplified versions of our models to provide efficient decision support for resource-limited emergency medical environments. These models facilitate early identification of TBI patients requiring surgical intervention and massive blood transfusion, while supporting rapid triage and blood allocation decisions in emergency medical settings (Fig. 1).

Results

Development and validation of the LR-based CBI model for TBI surgical indicators

The primary objective of this model was to predict the need for surgical intervention versus conservative treatment for patients with TBI upon hospital admission. We analyzed four clinical variables in the training set: pupillary response (categorized as bilateral normal pupils, bilateral constricted pupils, bilateral dilated pupils, or anisocoria), level of consciousness (conscious, drowsy, or comatose), age, and sex. Univariable logistic regression analysis revealed significant associations for pupillary response (OR: 1.40–10.51), level of consciousness (OR: 2.06–7.16), and age categories (OR: 2.27–2.42) ($p < 0.001$) (Fig. 2a). Multivariable logistic regression analysis confirmed these associations, revealing progressive increases in the odds ratio with age category (OR: 1.72–2.18), increased risk across consciousness levels (OR: 1.95–5.74), and significant risks for a partial pupillary response (OR: 0.74, 4.61, and 1.67, respectively) (Fig. 2b). Following the evaluation of eight ML algorithms in both the training set and internal set, we selected the LR model as the optimal approach on the basis of its superior discriminative capability while maintaining interpretability (Fig. 2c). A practical nomogram incorporating age, consciousness level, and pupillary response as key predictors was developed for use in clinical practice (Fig. 2d). The predictive performance was robust across the data from multiple centers, with AUC values consistently exceeding 0.70 (training set: 0.74; validation set: 0.76; external testing sets 1–3: 0.75) (Fig. 2e, Table 2). Calibration analysis revealed moderate concordance between the predicted and observed probabilities across all datasets (Fig. 2f). DCA demonstrated the clinical utility of the model, supporting evidence-based decision-making for the management of TBI patients (Fig. 2g).

Development and performance evaluation of the 2.5D deep transfer learning model for CT-based surgical planning

We developed a 2.5D NRI model utilizing deep transfer learning to evaluate surgical indicators in patients with TBI based on cranial CT scan data. The model was developed via an innovative multislice analysis strategy centered on the maximum cross-sectional area of the brain and a two-stage training strategy: the training set was divided into an initial training set and a refinement set for 2.5D deep learning model training and validation. The features extracted from the refinement set by the trained deep learning model were subsequently used in a second training phase involving multiple ML methods, and final validation was performed in the validation set (Fig. 3a). Following extensive comparative analysis of multiple deep learning architectures, MobileNet V2 emerged as the superior performer (Fig. 3b). Grad-CAM demonstrated remarkable spatial correspondence between the regions from which MobileNet V2 extracted key features and areas of cerebral hemorrhage, validating the selection of this model as the optimal feature extractor (Fig. 3c and detailed in Supplementary Fig. 2). In the feature engineering phase, we employed PCA to reduce the dimensionality of the extracted features to 36 components, followed by LASSO regression for refined feature selection (Fig. 3d). Among the evaluated ML algorithms, a transfer-trained LR model demonstrated superior performance (Fig. 3e). This model achieved impressive discriminative ability, with AUC values of 0.89 (95% CI: 0.84–0.93) in the refinement set and 0.83 (95% CI: 0.77–0.88) in the validation set.

While the model exhibited greater performance than the CBI model in the internal validation set did, it showed lower stability across the three external testing sets (AUC range: 0.70–0.81) (Fig. 3f). Calibration analysis revealed a tendency to underestimate surgical intervention requirements in

Table 1 | Demographic characteristics of all included patients and surgical patients in the study

	Variable	Overall	Training (Cohort 1)	Validation (Cohort 1)	External Testing 1 (Cohort 2)	External Testing 2 (Cohort 2)	External Testing 3 (Cohort 3)
		<i>N</i> = 2219 ¹	<i>N</i> = 848 ¹	<i>N</i> = 212 ¹	<i>N</i> = 501 ¹	<i>N</i> = 262 ¹	<i>N</i> = 396 ¹
For all included patients	Age (years)						
	Median (Q1–Q3)	52.00 (40.00–61.00)	55.00 (44.00–64.00)	54.00 (40.50–63.50)	51.00 (40.00–60.00)	54.00 (41.00–60.00)	52.00 (40.00–60.00)
	Sex, <i>n</i> (%)						
	Female	595 (27)	237 (28)	64 (30)	139 (28)	54 (21)	101 (26)
	Male	1,624 (73)	611 (72)	148 (70)	362 (72)	208 (79)	295 (74)
	Surgery, <i>n</i> (%)						
	No	1127 (51)	387 (46)	96 (45)	357 (71)	123 (47)	164 (41)
	Yes	1092 (49)	461 (54)	116 (55)	144 (29)	139 (53)	232 (59)
	Overall	Training (Cohort 1)	Validation (Cohort 1)	External Testing 1 (Cohort 2)	External Testing 2 (Cohort 2)	External Testing 3 (Cohort 3)	
	<i>N</i> = 1092 ¹	<i>N</i> = 461 ¹	<i>N</i> = 116 ¹	<i>N</i> = 144 ¹	<i>N</i> = 139 ¹	<i>N</i> = 232 ¹	
For all surgical patients	Age (years)						
	Median (Q1–Q3)	53.00 (43.00–62.00)	56.00 (46.00–64.00)	57.00 (43.00 – 65.500)	50.00 (40.00–59.50)	55.00 (47.00–61.00)	53.00 (43.00–61.00)
	Sex, <i>n</i> (%)						
	Female	279 (26)	125 (27)	37 (32)	31 (22)	29 (21)	57 (25)
	Male	813 (74)	336 (73)	79 (68)	113 (78)	110 (79)	175 (75)
	Transfusion, <i>n</i> (%)						
	No	640 (59)	281 (61)	70 (60)	66 (46)	85 (61)	138 (59)
	Yes	452 (41)	180 (39)	46 (40)	78 (54)	54 (39)	94 (41)
	Time from admission examination to surgery (h) ²						
Median (Q1–Q3)	3.00 (2.50–10.00)	3.00 (2.50–9.00)	4.00 (2.50–10.50)	3.00 (2.00–13.25)	3.00 (2.50–7.00)	3.00 (2.50–9.00)	

¹Median (IQR) or frequency (%).²The median time interval between completion of initial admission records (establishment of initial medical history, diagnosis, blood tests, and CT reports, enabling model-based prediction output) and surgery initiation.

external testing set 1 (Fig. 3g). Nevertheless, DCA confirmed the model's clinical utility across a broad range of threshold probabilities (Fig. 3h), suggesting that its potential value remains in clinical decision-making.

Development and validation of the CTM model for admission-based surgical decision support in TBI patients

In addition to the previously described models, we developed a comprehensive NLP approach, named the CTM model, to extract predictive features from admission texts for admission-based surgical decision support in TBI patients (Fig. 4a). The Boruta algorithm identified key clinical attributes, and feature importance analysis highlighted brain herniation, multiple cerebral contusions, and subarachnoid hemorrhage as the most significant predictive variables (Fig. 4b).

Among the eight ML algorithms evaluated in the various datasets, the LR model demonstrated consistently superior performance in both the training and validation sets, leading to its selection as the optimal model (Fig. 4c). The final multivariable LR model incorporated multiple clinical variables, among which brainstem injury and cerebral hemorrhage were identified as the two strongest predictors for surgical intervention according to coefficient analysis (Fig. 4d). The model achieved AUC values of 0.73 (95% CI: 0.70–0.77) in the training set and 0.71 (95% CI: 0.64–0.77) in the validation set and achieved stable performance across the three external validation sets, with AUC values ranging from 0.68 to 0.69 (Fig. 4e). Although the model exhibited relatively modest discriminative ability, its reliance solely on clinical text data without requiring examination results underscores its potential as an early screening tool. Calibration analysis revealed suboptimal calibration in most validation and external testing sets (Fig. 4f), whereas DCA confirmed the model's clinical utility across various threshold probabilities (Fig. 4g), suggesting its potential value as an early screening tool for TBI surgical decision support.

Development and validation of a CBI Model for Admission-based Transfusion Prediction in Surgical TBI Patients

The second objective of our predictive modeling focused on identifying the need for massive transfusion, defined as the administration of more than 4 units of RBCs, in surgical TBI patients. The model development process began with comprehensive feature selection from a pool of 146 clinical variables, encompassing five major categories: baseline characteristics, biochemical test results, coagulation test results, blood gas analysis results, and complete blood counts (Supplementary Table 1). We employed RFE to identify the 30 most significant predictive variables for subsequent analysis (Fig. 5A). Further refinement was achieved through LASSO regression with cross-validation, ultimately yielding nine key variables for construction of the final model: consciousness level, Lac (lactic acid), TP (total protein), AST (aspartate aminotransferase), K (potassium), GCS score (Glasgow Coma Scale score), RBC (red blood cell count), HGB (hemoglobin), and PLR (pupillary light reflex) (Fig. 5B). Among the eight ML algorithms evaluated, the XGB model demonstrated consistently superior performance in both the training and validation sets and was thus selected as the optimal model (Fig. 5c). SHAP analysis provided interpretable insights into feature importance, revealing that elevated lactate levels, decreased GCS scores, and low hemoglobin were the three most significant predictors of the need for transfusion (Fig. 5d).

The model exhibited robust performance in the training and validation sets, achieving AUC values of 0.86 and 0.82, respectively. However, upon evaluation of the merged external testing datasets, we observed a substantial decline in predictive ability (AUC = 0.69, 95% CI: 0.63–0.73). This marked decrease in performance indicates limited model generalizability, suggesting that in isolation, clinical phenotypic data are insufficient for comprehensively assessing the need for intraoperative transfusion in TBI patients (Fig. 5e). Calibration analyses demonstrated that the CBI model systematically

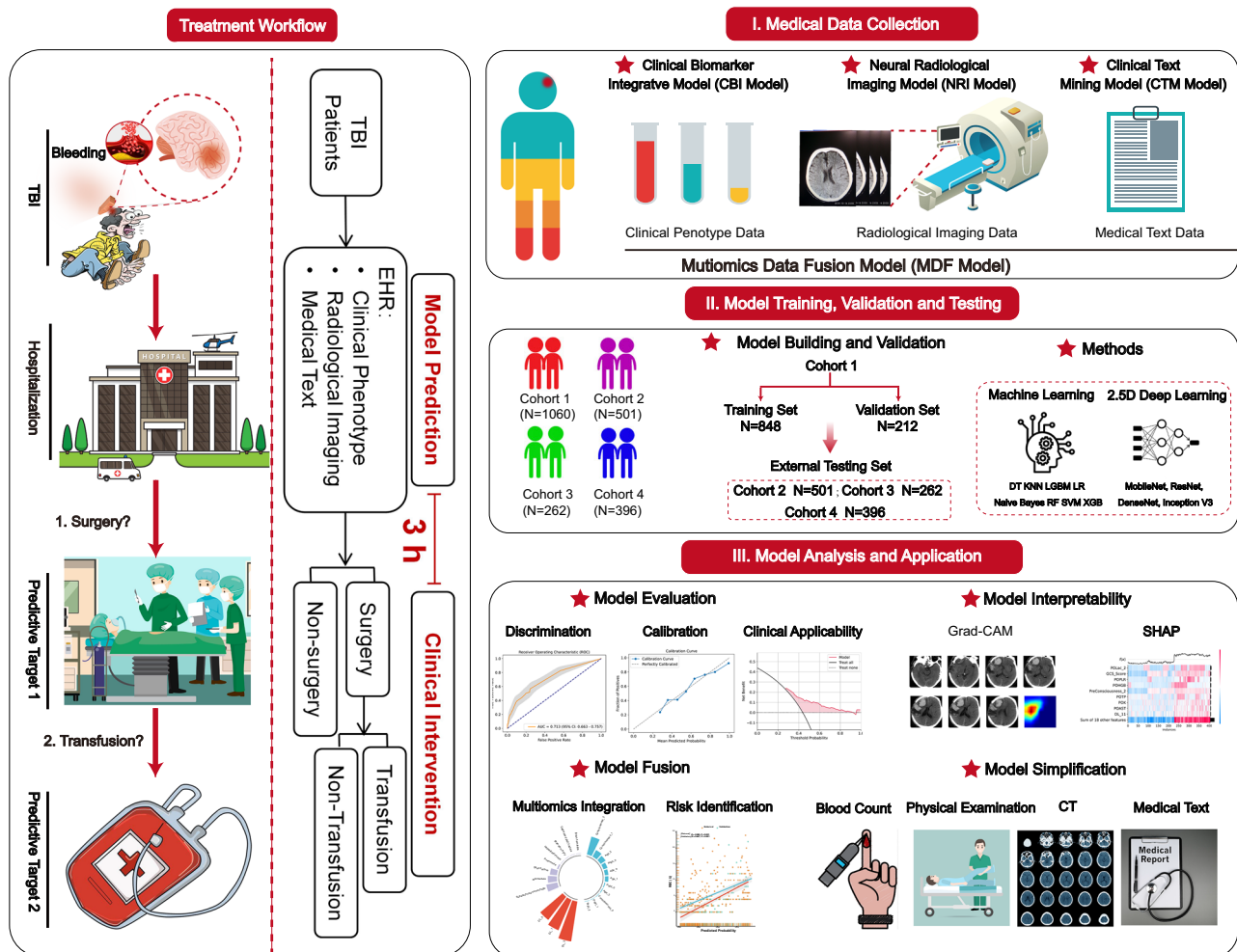


Fig. 1 | Schematic showing the workflow of the ML-based pipeline to predict indicators for surgery and blood transfusion. The figure illustrates the comprehensive workflow of models developed in this study, comprising three distinct

phases: I. Medical Data Collection, II. Model Training, Validation and Testing, and III. Model Analysis and Clinical Application.

overestimated the need for transfusion across the external testing cohorts (Fig. 5f). Furthermore, DCA revealed limited clinical utility in the external testing sets (Fig. 5g). Taken together, these findings demonstrate the substantial challenges inherent in developing standardized transfusion risk prediction tools that are both consistent and reliable across diverse institutions.

Development of ML models based on radiological and clinical text data for admission-based transfusion prediction in TBI patients

Similarly, we developed two ML models based on radiological and clinical text data from admission records for the early prediction of intra/post-operative transfusion (>4 U RBCs) in TBI patients. For the radiological-based model (NRI model), RFE was implemented to select significant imaging-derived features (Fig. 6a). An evaluation of eight ML algorithms across both the training and validation sets identified RF as the optimal model (Fig. 6b). External validation revealed an AUC of 0.73 (95% CI: 0.68–0.78), marginally exceeding the value for the CBI Model (Fig. 6c and Supplementary Fig. 3; details on the performance in the training and validation sets are presented in Supplementary Fig. 1).

For the CTM model, the Boruta algorithm was employed for feature selection from the structured admission text data (Fig. 6d), followed by a comprehensive evaluation of eight ML algorithms (Fig. 6e). The final model was the LR model, and brainstem injury,

pulmonary edema, and sacroiliac joint fracture were identified as the three strongest predictors of the need for transfusion (Fig. 6f). This text-based model achieved an AUC of 0.69 (95% CI: 0.64–0.74) in the external validation set (Fig. 6g and Supplementary Fig. 4; details on the performance in the training and validation sets are shown in Supplementary Fig. 1).

Interestingly, both models exhibited limitations similar to those of the CBI model during external validation, including suboptimal predictive performance, systematic overestimation of transfusion risk in the calibration analyses and limited clinical utility according to the DCA. These findings demonstrate the substantial challenges inherent in developing standardized transfusion prediction methods across diverse institutions while also suggesting that single-modality data sources, whether radiological, textual, or biochemical, may be insufficient for comprehensively assessing the need for intraoperative transfusion in TBI patients. These limitations emphasize the potential need for integrated, multimodal approaches in future predictive model development.

Development and performance evaluation of multimomics data fusion models (MDF Models) for surgery and transfusion prediction in TBI

The integration of multimomics data yielded two comprehensive predictive models following feature selection through LASSO regression

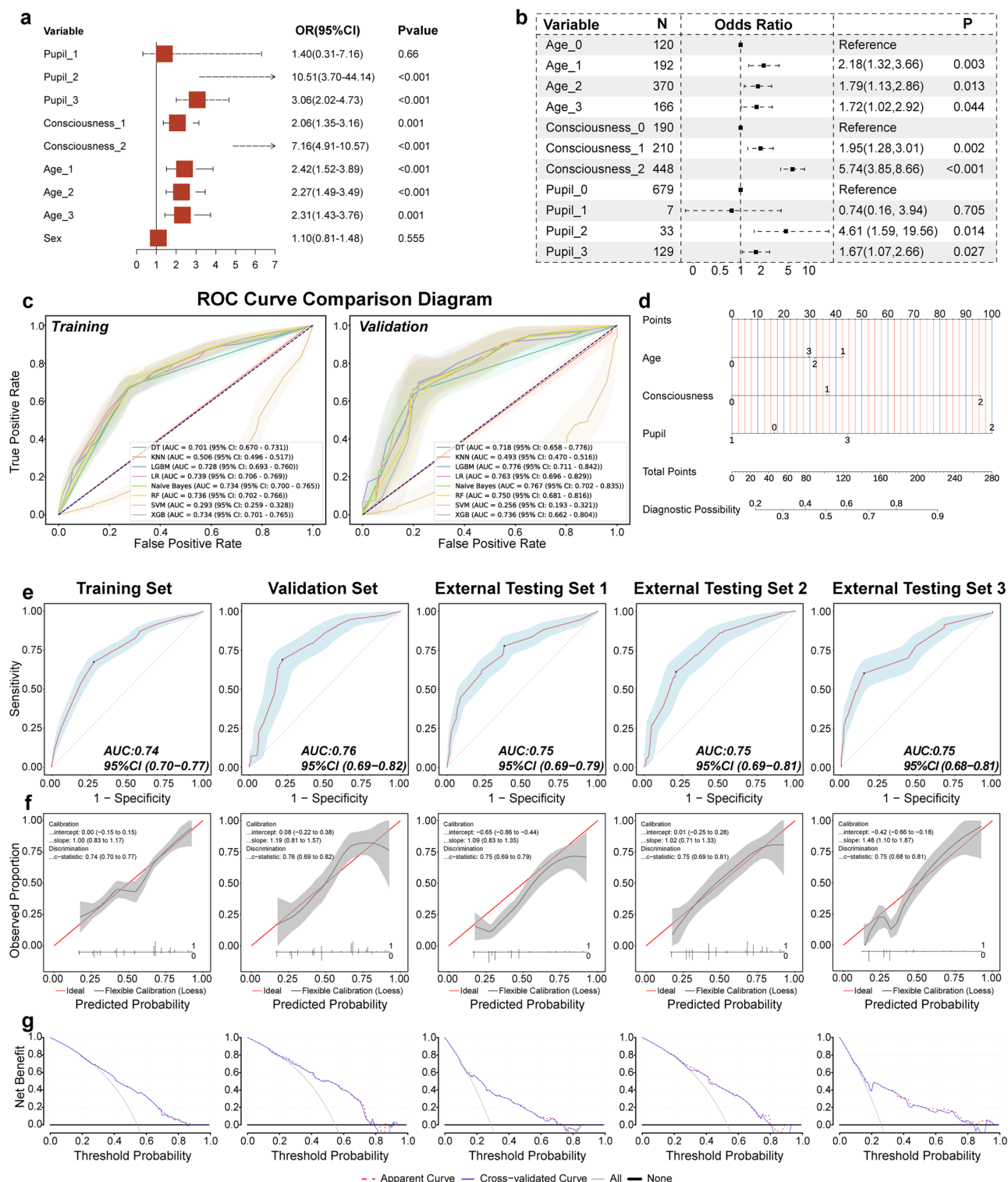


Fig. 2 | CBI model for surgery prediction in TBI patients. Initial variable screening through univariable analysis **a** followed by multivariable logistic regression **b** identified key predictors, including age, consciousness level, and pupillary response. Model performance was evaluated via various ML approaches (**c**) and visualized with a practical nomogram (**d**). Extensive validation was performed across multiple datasets (**e**), while model calibration assessment (**f**) and decision curve analysis (**g**) demonstrated the clinical utility of the model. Age_0: <35 years;

Age_1: 35-50 years; Age_2: 50-65 years; Age_3: >65 years; Consciousness_0: Conscious; Consciousness_1: Drowsy; Consciousness_2: Comatose; Pupil_0: Bilateral Normal Pupils; Pupil_1: Bilateral Constricted Pupils; Pupil_2: Bilateral Dilated Pupils; Pupil_3: Anisocoria; DT: Decision Tree; KNN: K-nearest neighbors; LGBM: Light gradient boosting machine; LR: Logistic Regression; RF: Random Forest; SVM: Support Vector Machine; XGB: eXtreme Gradient Boosting.

Table 2 | Summary of the predictive performance of all the models

Model name	Cohort	AUC (95% CI)	BrierScore (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)	Accuracy (95%CI)	F1 score (95% CI)	Sample size (95% CI)	Model architecture
Clinical biomarker integrative model for surgery	Training	0.74 (0.7–0.77)	0.21 (0.19–0.22)	0.67 (0.63–0.72)	0.72 (0.68–0.78)	0.7 (0.67–0.73)	0.71 (0.67–0.74)	848	LR
	Validation	0.76 (0.69–0.82)	0.19 (0.17–0.22)	0.69 (0.6–0.82)	0.78 (0.64–0.88)	0.73 (0.68–0.79)	0.74 (0.67–0.8)	212	
	External Testing 1	0.75 (0.69–0.79)	0.18 (0.17–0.2)	0.78 (0.46–0.84)	0.62 (0.58–0.92)	0.66 (0.64–0.8)	0.57 (0.52–0.64)	501	
	External Testing 2	0.75 (0.69–0.81)	0.2 (0.18–0.23)	0.61 (0.55–0.89)	0.79 (0.49–0.86)	0.69 (0.65–0.76)	0.68 (0.63–0.78)	262	
	External Testing 3	0.75 (0.68–0.81)	0.16 (0.14–0.17)	0.6 (0.5–0.69)	0.85 (0.81–0.91)	0.79 (0.75–0.83)	0.59 (0.52–0.67)	396	
	Refinement	0.89 (0.84–0.93)	0.13 (0.11–0.16)	0.89 (0.72–0.94)	0.77 (0.7–0.93)	0.83 (0.79–0.89)	0.85 (0.79–0.9)	210	
Neural radiological imaging model for surgery	Validation	0.83 (0.77–0.88)	0.17 (0.14–0.21)	0.77 (0.53–0.93)	0.74 (0.56–0.96)	0.75 (0.7–0.82)	0.77 (0.67–0.84)	212	MobileNet V2 + LR
	External Testing 1	0.7 (0.64–0.75)	0.25 (0.22–0.28)	0.78 (0.65–0.85)	0.58 (0.52–0.7)	0.64 (0.6–0.7)	0.57 (0.51–0.63)	429	
	External Testing 2	0.78 (0.71–0.83)	0.21 (0.17–0.24)	0.83 (0.65–0.89)	0.66 (0.59–0.83)	0.75 (0.7–0.8)	0.77 (0.7–0.82)	248	
	External Testing 3	0.81 (0.77–0.86)	0.18 (0.15–0.2)	0.81 (0.66–0.9)	0.7 (0.6–0.85)	0.77 (0.72–0.83)	0.82 (0.75–0.87)	367	
	Training	0.73 (0.7–0.77)	0.2 (0.19–0.21)	0.66 (0.58–0.7)	0.72 (0.68–0.8)	0.69 (0.66–0.72)	0.69 (0.65–0.73)	848	
	Validation	0.71 (0.64–0.77)	0.21 (0.19–0.24)	0.72 (0.3–0.81)	0.6 (0.55–0.99)	0.67 (0.59–0.74)	0.71 (0.45–0.77)	212	
Clinical text mining model for surgery	External Testing 1	0.68 (0.62–0.74)	0.24 (0.22–0.25)	0.38 (0.3–0.71)	0.92 (0.63–0.97)	0.77 (0.63–0.81)	0.48 (0.42–0.57)	501	XGB
	External Testing 2	0.68 (0.61–0.75)	0.25 (0.22–0.27)	0.79 (0.59–0.86)	0.63 (0.55–0.81)	0.71 (0.66–0.77)	0.75 (0.66–0.8)	262	
	External Testing 3	0.69 (0.64–0.74)	0.22 (0.2–0.24)	0.72 (0.59–0.78)	0.61 (0.54–0.75)	0.68 (0.63–0.73)	0.74 (0.67–0.78)	396	
	Refinement	0.95 (0.92–0.98)	0.09 (0.07–0.11)	0.88 (0.8–0.95)	0.92 (0.85–0.99)	0.9 (0.87–0.94)	0.9 (0.87–0.95)	210	
	Validation	0.87 (0.82–0.91)	0.15 (0.12–0.18)	0.79 (0.71–0.88)	0.82 (0.73–0.91)	0.81 (0.76–0.86)	0.82 (0.76–0.88)	212	
	External Testing 1	0.81 (0.76–0.86)	0.17 (0.16–0.18)	0.81 (0.5–0.88)	0.67 (0.63–)	0.72 (0.69–0.85)	0.63 (0.59–0.71)	429	
Mutimomics data fusion model for surgery	External Testing 2	0.81 (0.75–0.86)	0.18 (0.16–0.21)	0.8 (0.68–0.88)	0.72 (0.66–0.86)	0.77 (0.72–0.82)	0.78 (0.72–0.84)	248	XGB
	External Testing 3	0.87 (0.83–0.91)	0.15 (0.13–0.17)	0.86 (0.65–0.94)	0.73 (0.65–0.93)	0.81 (0.74–0.86)	0.85 (0.77–0.89)	367	
	Training	0.86 (0.83–0.9)	0.15 (0.13–0.17)	0.81 (0.65–0.85)	0.77 (0.74–0.93)	0.79 (0.76–0.84)	0.77 (0.72–0.82)	412	
	Validation	0.82 (0.73–0.89)	0.17 (0.13–0.22)–	0.74 (0.57–0.86)	0.85 (0.76–0.95)	0.8 (0.73–0.88)	0.76 (0.66–0.86)	106	
	Merged External Testing	0.69 (0.63–0.73)	0.24 (0.23–0.25)	0.65 (0.6–0.74)	0.67 (0.57–0.74)	0.66 (0.62–0.71)	0.67 (0.62–0.72)	436	
	Training	1.0 (1.0–1.0)	0.04 (0.03–0.04)	1.0 (1.0–1.0)	1.0 (1.0–1.0)	1.0 (1.0–1.0)	1.0 (1.0–1.0)	406	
Neural radiological imaging model for blood transfusion	Validation	0.74 (0.66–0.83)	0.21 (0.19–0.23)	0.78 (0.54–0.93)	0.63 (0.48–0.88)	0.7 (0.63–0.8)	0.69 (0.59–0.8)	106	RF
	Merged External Testing	0.73 (0.68–0.78)	0.21 (0.19–0.22)	0.62 (0.37–0.98)	0.71 (0.33–0.95)	0.66 (0.63–0.71)	0.65 (0.52–0.76)	396	
	Training	0.71 (0.66–0.76)	0.21 (0.2–0.23)	0.57 (0.37–0.84)	0.75 (0.46–0.92)	0.67 (0.62–0.72)	0.6 (0.49–0.67)	412	
	Validation	0.68 (0.58–0.77)	0.22 (0.2–0.26)	0.59 (0.43–0.88)	0.75 (0.43–0.87)	0.68 (0.58–0.76)	0.61 (0.49–0.73)	106	
	Merged External Testing	0.69 (0.64–0.74)	0.24 (0.22–0.25)	0.5 (0.44–0.89)	0.79 (0.39–0.85)	0.64 (0.61–0.69)	0.59 (0.54–0.75)	436	
	Training	1.0 (1.0–1.0)	0.03 (0.03–0.03)	1.0 (1.0–1.0)	1.0 (1.0–1.0)	1.0 (1.0–1.0)	1.0 (1.0–1.0)	406	
Multi-omics data fusion model for blood transfusion	Validation	0.88 (0.81–0.94)	0.15 (0.12–0.17)	0.67 (0.6–0.96)	0.95 (0.66–1.0)	0.83 (0.76–0.9)	0.78 (0.71–0.88)	106	RF
	Merged External Testing	0.82 (0.78–0.86)	0.19 (0.17–0.22)	0.65 (0.6–0.78)	0.88 (0.75–0.92)	0.77 (0.73–0.81)	0.74 (0.69–0.79)	396	
	Training	0.84 (0.8–0.88)	0.18 (0.17–0.19)	0.8 (0.61–0.86)	0.73 (0.68–0.9)	0.76 (0.73–0.81)	0.74 (0.68–0.79)	406	
	Validation	0.81 (0.73–0.89)	0.17 (0.13–0.21)	0.63 (0.52–0.9)	0.9 (0.65–0.97)	0.78 (0.72–0.87)	0.72 (0.64–0.83)	106	
	Merged External Testing	0.75 (0.7–0.8)	0.21 (0.2–0.23)	0.55 (0.45–0.9)	0.83 (0.46–0.92)	0.69 (0.65–0.73)	0.64 (0.57–0.76)	396	
	Training	0.84 (0.8–0.88)	0.18 (0.17–0.19)	0.8 (0.61–0.86)	0.73 (0.68–0.9)	0.76 (0.73–0.81)	0.74 (0.68–0.79)	406	
Simplified multi-omics data fusion model for blood transfusion	Validation	0.81 (0.73–0.89)	0.17 (0.13–0.21)	0.63 (0.52–0.9)	0.9 (0.65–0.97)	0.78 (0.72–0.87)	0.72 (0.64–0.83)	106	XGB
	Merged External Testing	0.75 (0.7–0.8)	0.21 (0.2–0.23)	0.55 (0.45–0.9)	0.83 (0.46–0.92)	0.69 (0.65–0.73)	0.64 (0.57–0.76)	396	
	Training	0.84 (0.8–0.88)	0.18 (0.17–0.19)	0.8 (0.61–0.86)	0.73 (0.68–0.9)	0.76 (0.73–0.81)	0.74 (0.68–0.79)	406	
	Validation	0.81 (0.73–0.89)	0.17 (0.13–0.21)	0.63 (0.52–0.9)	0.9 (0.65–0.97)	0.78 (0.72–0.87)	0.72 (0.64–0.83)	106	
	Merged External Testing	0.75 (0.7–0.8)	0.21 (0.2–0.23)	0.55 (0.45–0.9)	0.83 (0.46–0.92)	0.69 (0.65–0.73)	0.64 (0.57–0.76)	396	
	Training	0.84 (0.8–0.88)	0.18 (0.17–0.19)	0.8 (0.61–0.86)	0.73 (0.68–0.9)	0.76 (0.73–0.81)	0.74 (0.68–0.79)	406	

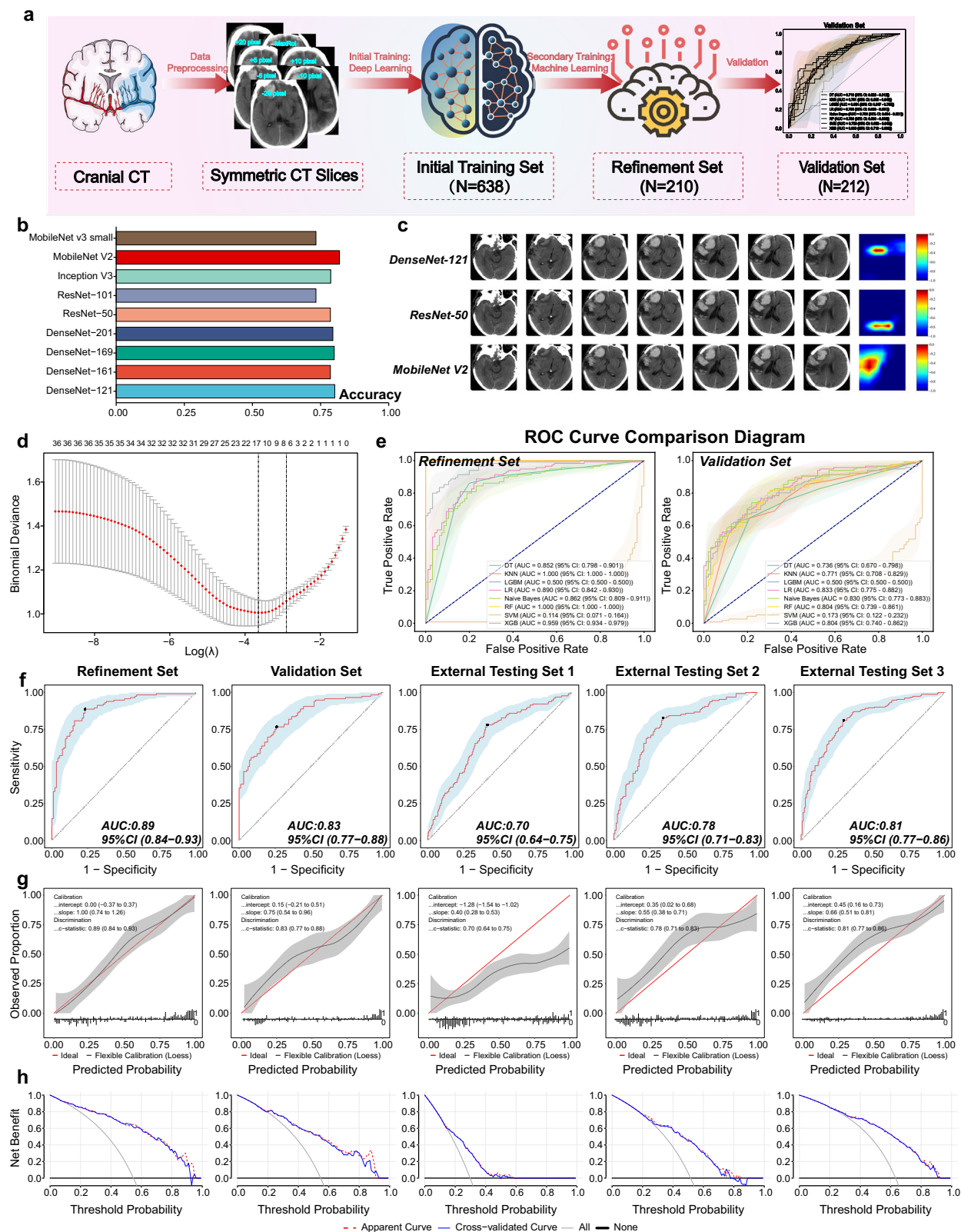


Fig. 3 | NRI Model for Surgery Prediction in TBI Patients. **a** Overview of the 2.5D deep learning framework: Seven strategic slices (central slices ± 5 , ± 10 , ± 20 pixels) from the patient's brain CT image are processed to form a 7-channel input for deep learning and ML analysis. **b** Comparative performance of different deep learning architectures, including MobileNet, Inception, ResNet, and DenseNet variants. **c** Grad-CAM visualization results for DenseNet-121, ResNet-50, and MobileNet V2, demonstrating image areas where the models focused their attention. **d** LASSO

regression for feature selection optimization. **e** ROC curves demonstrating the performance of multiple ML classifiers. **f** Model performance across the refinement set (AUC = 0.89), validation set (AUC = 0.83), and three external testing sets (AUC = 0.70-0.82). **g** Calibration plots showing predicted versus observed probabilities across all datasets. **h** Decision curve analysis demonstrating the clinical utility across various threshold probabilities. MaxRoI Max Region of interest.

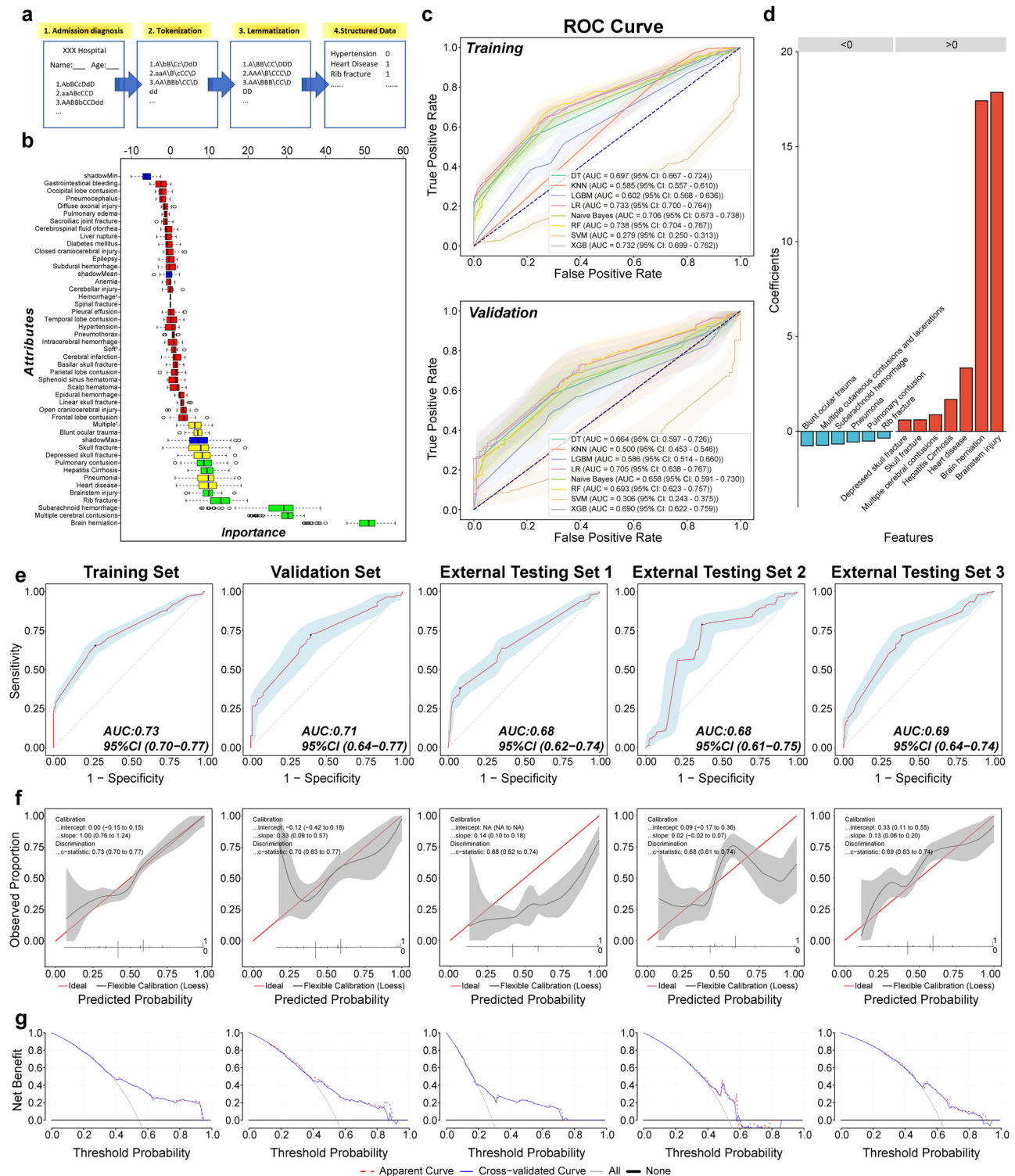
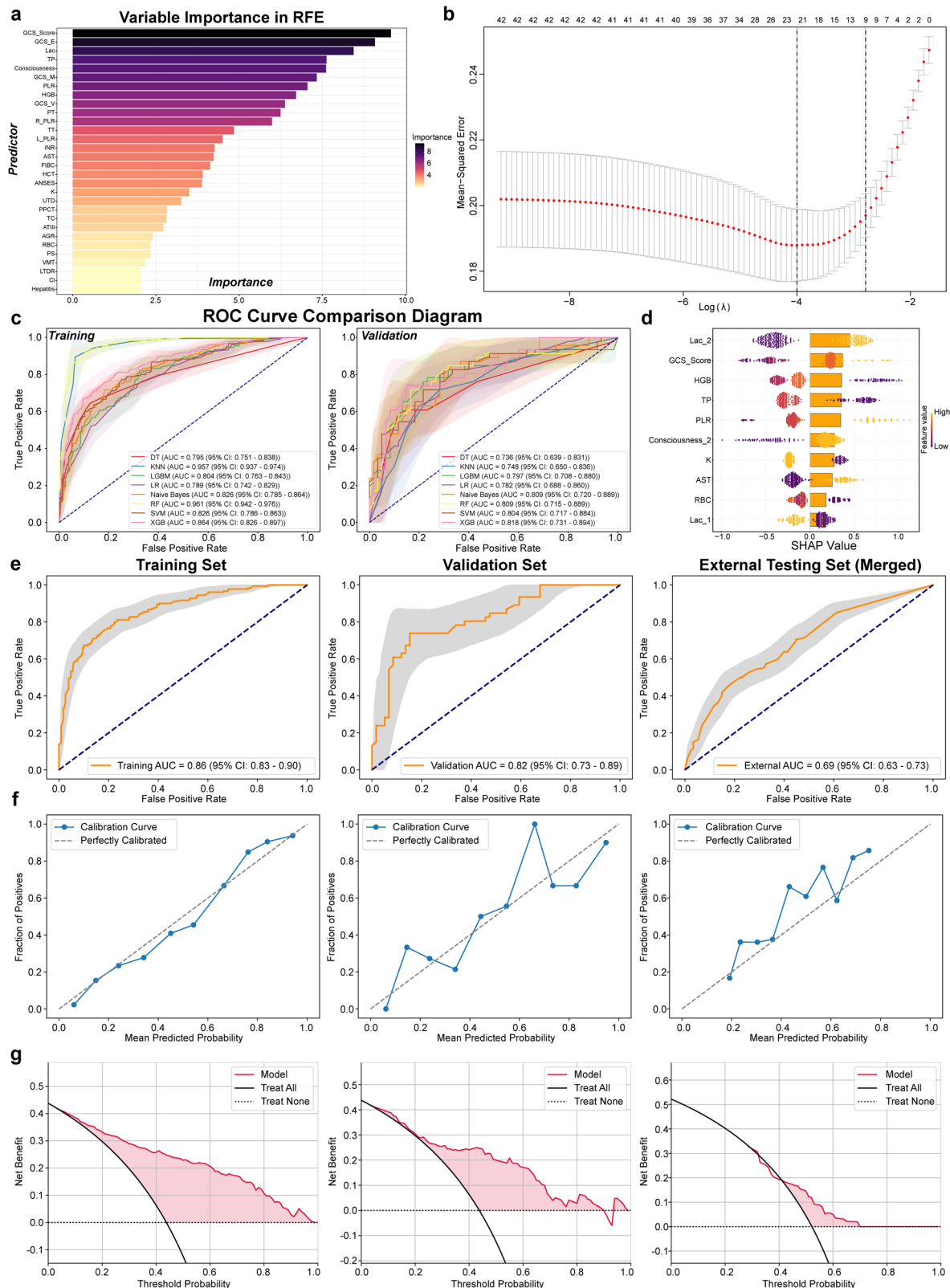


Fig. 4 | CTM Model for Surgery Prediction in TBI Patients. **a** Model development pipeline demonstrating the conversion of admissions text into structured data, involving steps including tokenization and lemmatization. **b** Feature selection via the Boruta algorithm; the panel shows the importance of various attributes used in the model. **c** Training and validation results of eight ML algorithms applied to the training and validation cohorts. **d** Variables included in the final multivariable LR model; the coefficients indicate the influence of each variable on the prediction.

Model performance evaluation across all datasets (training, validation, and three external testing sets) via **e** receiver operating characteristic (ROC) curves for discriminative ability, **f** calibration plots for prediction accuracy, and **g** decision curve analysis for clinical utility assessment. ^a: Hemorrhage from multiple cerebral contusions; ^b: Soft tissue swelling of the head; ^c: Multiple cutaneous contusions and lacerations.



(Supplementary Fig. 5a, b). The surgical indicator model incorporated 14 key features, including clinical phenotypic features (consciousness level, age, and pupil status), radiological imaging features (three deep learning features), and information extracted from medical texts (including subarachnoid hemorrhage, fracture patterns, and several complications). The transfusion prediction model included 23

features, including laboratory parameters, clinical scores, deep learning-derived imaging features, and text-mined clinical complications. XGB demonstrated optimal performance for surgical prediction across all cohorts (Fig. 7a), whereas RF emerged as the superior algorithm for transfusion prediction (Fig. 7b). Notably, both MDF models exhibited greater discriminative ability and generalizability

Fig. 5 | CBI model for transfusion prediction in TBI patients. **a** Initial feature selection via RFE from 146 clinical variables; the panel shows variable importance rankings for baseline characteristics, biochemical test parameters, coagulation test parameters, blood gas analysis parameters, and complete blood counts. **b** LASSO regression results demonstrating refined feature selection with optimal lambda value selection through cross-validation. **c** Comparison of the performance of eight ML algorithms in the training and validation cohorts, displaying ROC curves with corresponding AUC values for each model. **d** SHAP analysis of the XGBoost model, revealing the contribution and importance ranking of each selected feature in

predicting the need for transfusion. Model performance evaluation across all datasets (training, validation, and merged external test sets) via **e** receiver operating characteristic (ROC) curves for discriminative ability, **f** calibration plots for prediction accuracy, and **g** decision curve analysis for clinical utility assessment. Consciousness_0: conscious; Consciousness_1: drowsy; Consciousness_2: comatose; Lac: lactic acid; TP: total protein; AST: aspartate aminotransferase; K: potassium ion; GCS score: Glasgow Coma Scale score, GCS score_0: > 15; GCS score_1: 15--13; GCS score_2: 13--9; GCS score_3: 9--3; GCS score_4: < 3; RBC red blood cell count, HGB hemoglobin, PLR pupillary light reflex.

than the corresponding single-domain approaches did, as evidenced by ROC curve analysis, calibration plots, and DCA (Supplementary Fig. 5c–e for the surgery indicator model and Supplementary Fig. 5f–h for the transfusion model).

To evaluate potential demographic biases in model performance, we conducted comprehensive subgroup analyses stratified by sex and age across all the external validation datasets (Supplementary Fig. 6). The MDF models maintained consistent performance across most demographic subgroups, with no significant degradation in predictive ability observed between male and female patients or across different age categories. This stability in performance across diverse patient demographics supports the fairness and generalizability of our models, although it should be noted that certain subgroups with limited sample sizes or single-category outcomes could not undergo complete model validation (as indicated in the ROC curve annotations).

To compare the heterogeneity observed in single-omics transfusion models with that observed in multiomics approaches, we employed rain cloud plots to analyze the prediction probability distributions for the transfused and nontransfused patients across the internal and external testing sets (Fig. 7c, d). The results showed that the CBI model and CTM model, which are predominantly based on discrete variables, exhibited discontinuous probability distributions with limited discriminative ability. Both the NRI models and the MDF models demonstrated robust discrimination in the internal test set. Notably, the MDF models maintained superior performance in the external testing sets, indicating greater robustness and generalizability.

This conclusion was further validated by radar plots of F1 scores (Fig. 7e, f and details in Table 2). In both the surgical indication and blood transfusion prediction tasks, the MDF model demonstrated consistently superior balanced performance across the validation and external testing cohorts. For surgical indication prediction, the model achieved an impressive F1 score of 0.82 in the validation cohort and maintained robust performance (F1: 0.63–0.85) across external testing cohorts, substantially outperforming other single-domain approaches (CBI: 0.57–0.68, NRI: 0.57–0.82, CTM: 0.48–0.75). Similarly, in terms of blood transfusion prediction, the multiomics model achieved the highest F1 scores in both the validation (0.78) and external testing (0.74) cohorts, which consistently surpassed those of the CBI (0.76, 0.67), NRI (0.69, 0.65), and CTM (0.61, 0.59) models in the respective cohorts. The superior F1 scores across both tasks reflect the model's robust ability to balance sensitivity and specificity, which was further supported by higher accuracy rates (surgical: 0.72–0.81; transfusion: 0.77–0.83) and better calibration, as indicated by lower Brier scores. This consistent performance advantage demonstrates the exceptional ability of the MDF model to integrate diverse data types and maintain reliable predictions across different clinical settings and prediction tasks, representing a significant advancement in multimodal clinical decision support systems.

Feature importance analysis and clinical validation of the MDF models in TBI management

To improve the possibility of the clinical implementation of our models, we conducted comprehensive model interpretability analyses. SHAP analysis revealed distinct feature importance patterns for the prediction of surgical

intervention (Fig. 8a) and blood transfusion (Fig. 8b). For surgical prediction, deep learning features (DL_1–3) derived from radiological imaging demonstrated the highest predictive weights, followed by the level of consciousness from among the clinical biomarkers and subarachnoid hemorrhage and skull fracture from among the medical text data. For transfusion prediction, clinical biomarkers, particularly high Lac levels, high GCS scores, high PLRs, and high HGB values, had the greatest importance weights. Radiological imaging-derived deep learning features showed moderate importance, whereas medical text data features, including rib fracture and multiple cerebral contusions, demonstrated lower predictive weights.

We further validated the clinical utility of the MDF model through systematic patient risk stratification. TBI patients who underwent surgical intervention were stratified into three risk categories (low, medium, and high) on the basis of the model-predicted transfusion risk. Analysis of the actual red blood cell transfusion volumes during perioperative care revealed statistically significant differences among the risk groups ($P < 0.05$ or 0.001) across both the internal and external testing sets (Fig. 8c). The model's predictive accuracy was further supported by correlation analysis between the predicted probabilities and the actual RBC transfusion volumes, which revealed strong associations in both the validation ($R = 0.687$, $P < 0.001$) and merged external testing sets ($R = 0.580$, $P < 0.001$) (Fig. 8d). These findings validate the model's potential as a clinical transfusion risk score, demonstrating its effectiveness in patient stratification and transfusion requirement prediction and supporting its potential integration into clinical decision support systems. To achieve this goal, we propose a workflow designed for deployment within real-world hospital settings, comprising three key components: clinical workflow integration, user training outline, and specific case demonstrations (Supplementary Notes 1–3).

Performance evaluation and clinical validation of the simplified multiomics prediction model

The complex biochemical and blood gas analysis data requirements of the comprehensive MDF model could limit its application in emergency situations. Therefore, a simplified version was developed that incorporates only routine complete blood count parameters, physical examination data, CT imaging features, and admission diagnostic text data. A systematic evaluation of multiple ML algorithms revealed that XGB had the best performance; therefore, this algorithm was selected as the optimal modeling approach (Fig. 9a). The model achieved robust performance in the training set (AUC: 0.84, 95% CI: 0.80–0.88) and demonstrated strong generalizability in both the validation and external testing sets (0.81, 95% CI: 0.73–0.89; and 0.75, 95% CI: 0.70–0.80, respectively). The calibration curves revealed moderate agreement between the predicted probabilities and observed outcomes (Fig. 9c), whereas DCA demonstrated that the model offered positive net clinical benefits (Fig. 9d). SHAP analysis identified key predictive factors: HGB from complete blood counts, five deep learning features (DL_17, DL_11, DL_30, DL_0, DL_23), and the GCS score and the PLR from the physical examination, thus validating the clinical utility of the model.

Discussion

This study developed a multi-omics fusion model integrating clinical phenotypic data, radiological imaging, and medical text data to predict surgical indications and blood transfusion requirements in traumatic brain injury

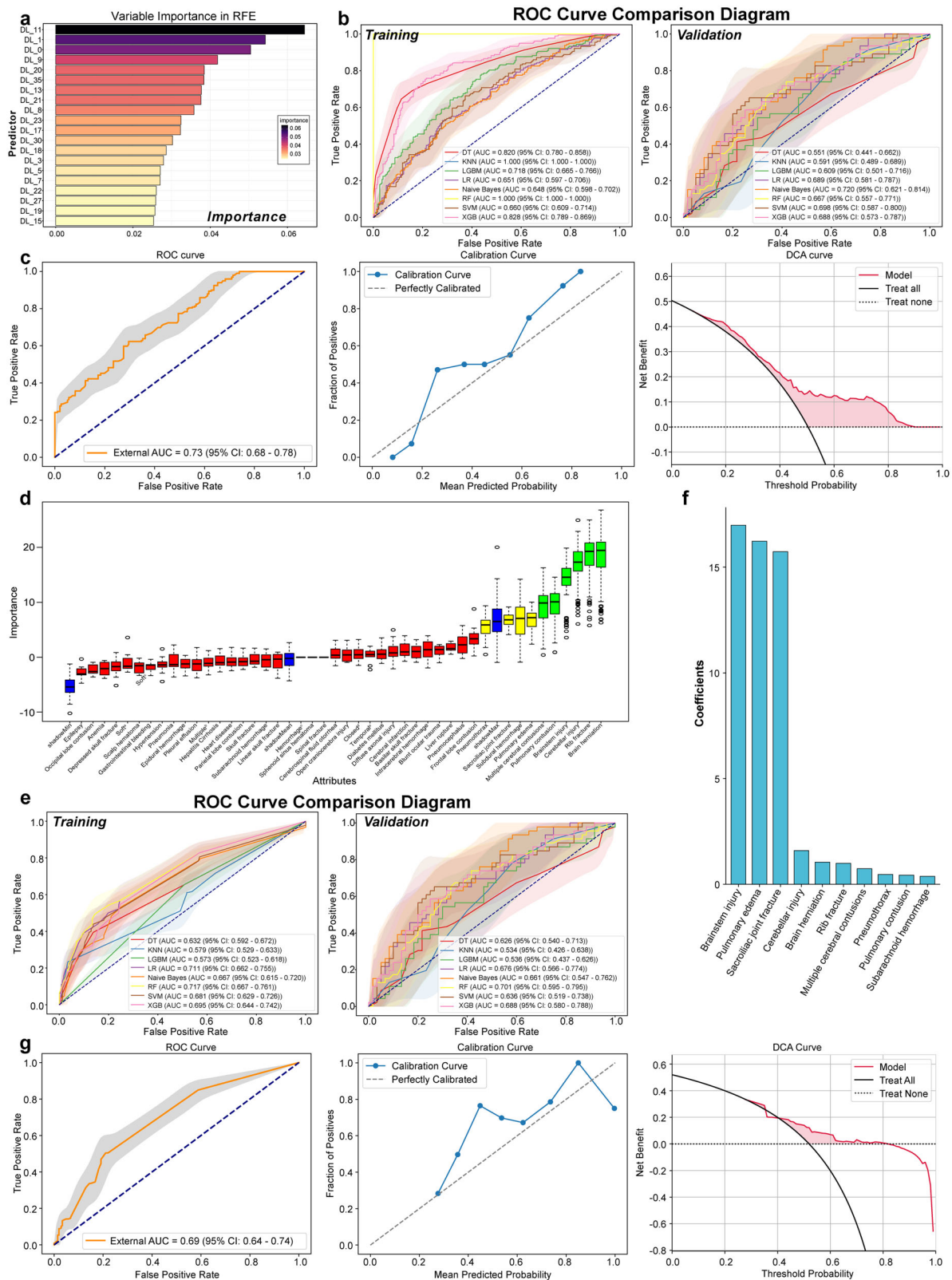


Fig. 6 | NRI and CTM models for fusion prediction in TBI patients. **a** Feature selection process employing RFE for deep learning features extracted from radiological images. **b** Comparison of the performance of eight ML algorithms in the training and validation sets, displaying ROC curves with corresponding AUC values for each model. **c** Performance evaluation of the optimized RF model in the external testing set. **d** Feature selection of structured data from the CTM model via the Boruta

algorithm for model construction. **e** Comparison of the performance of eight ML algorithms in the training and validation sets. **f** Features and their coefficients were incorporated into the optimal LR model. **g** Performance evaluation of the optimized LR model in the external testing set. ^a: Soft tissue swelling of the head; ^b: Multiple cutaneous contusions and lacerations; ^c: Hemorrhage from multiple cerebral contusions; ^d: Closed craniocerebral injury; ^e: Temporal lobe contusion.

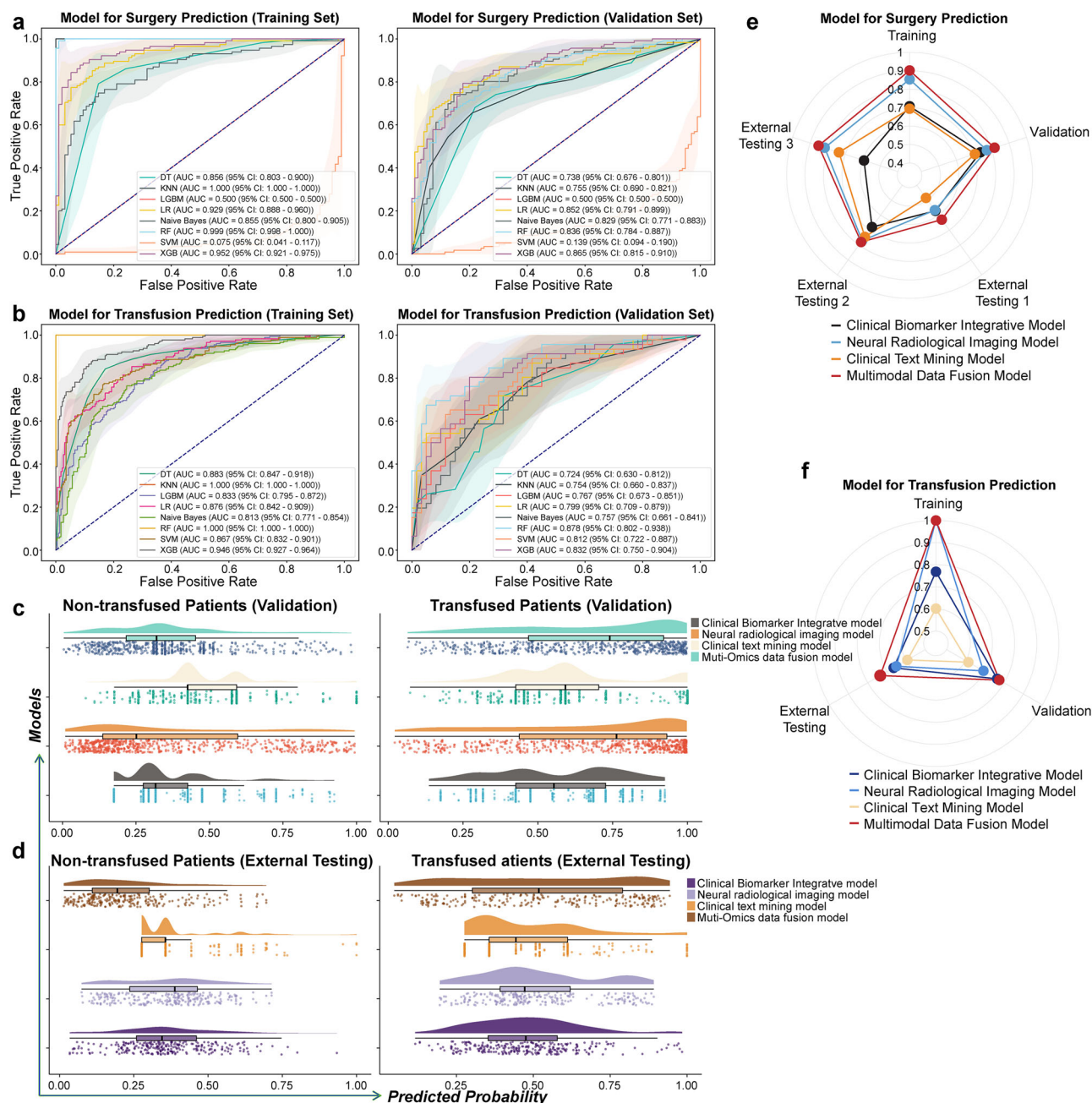


Fig. 7 | Multiomics data fusion model for surgery and transfusion prediction in TBI patients. A comprehensive multiomics approach that integrates biochemical markers, radiological imaging features, and CTM data was developed and validated for the prediction of the need for surgery and blood transfusion in traumatic brain injury (TBI) patients. **a, b** Receiver operating characteristic (ROC) curves illustrating the systematic evaluation of various ML algorithms during the model training and validation phases, utilizing carefully selected features from each data modality.

c, d Rain cloud plot visualization and probability density distributions comparing four distinct omics approaches (CBI, NRI, CTM, and MDF models), illustrating model discrimination between transfusion/nontransfusion patients across validation cohorts. **e, f** Comprehensive performance evaluation through radar plot visualization of F1 scores (which harmonically balance precision and recall) across multiple datasets, demonstrating the predictive accuracy for both surgical intervention indicators and the need for transfusion.

(TBI) patients. The model was validated using data from four independent clinical cohorts. All models—including both single-omics and multi-omics approaches—were rigorously developed and validated following TRI-POD + AI guidelines⁴³, with comprehensive evaluation of discrimination, calibration, and clinical utility (see Supplementary Table 2). By integrating explainable machine learning techniques such as SHAP and Grad-CAM, the model's decision-making process became transparent, enhancing its clinical credibility. It must be acknowledged that the multi-omics model did not perform best across all evaluation metrics, and some “less data intensive” models^{16,18} may perform better on specific indicators. However, our

comprehensive evaluation demonstrated that the multi-omics model exhibited the most consistent robustness in cross-center validation (see Table 2). Even under broad inclusion criteria—encompassing a wide age range, minimal restrictions on comorbidities, and no time limitations between admission and surgery—our constructed model maintained stable predictive performance, confirming its practical value. The end-to-end modeling strategy requires only standard admission data without additional examinations, thereby ensuring feasibility in clinical practice. Our research findings suggest that this model has the potential to support surgical decision-making assistance and may provide additional preparation time

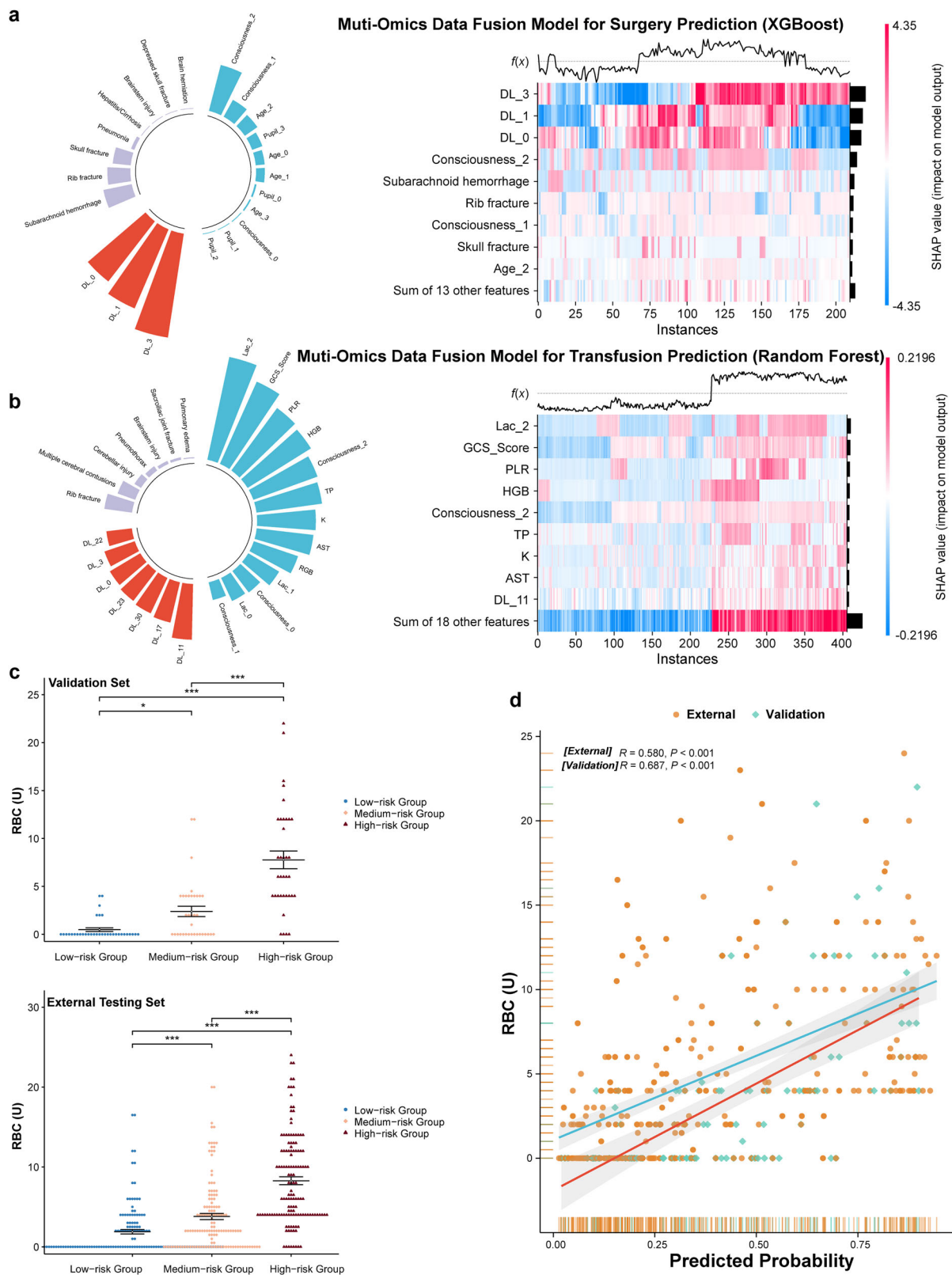


Fig. 8 | Feature importance analysis and clinical validation of the MDF Model. SHAP analysis showing feature importance weights for surgical intervention **a** and blood transfusion prediction **b** in circular plots and heatmap visualization of SHAP value distributions for key predictive features; $f(x)$ curves show the model output distribution. **c** Comparison of actual RBC transfusion volumes across risk groups

(low-, medium-, and high-risk) in the validation (upper) and external (lower) testing sets (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$). **d** Correlation analysis between the predicted transfusion probabilities and actual RBC transfusion volumes (internal: $R = 0.687$, $P < 0.001$; external: $R = 0.580$, $P < 0.001$).

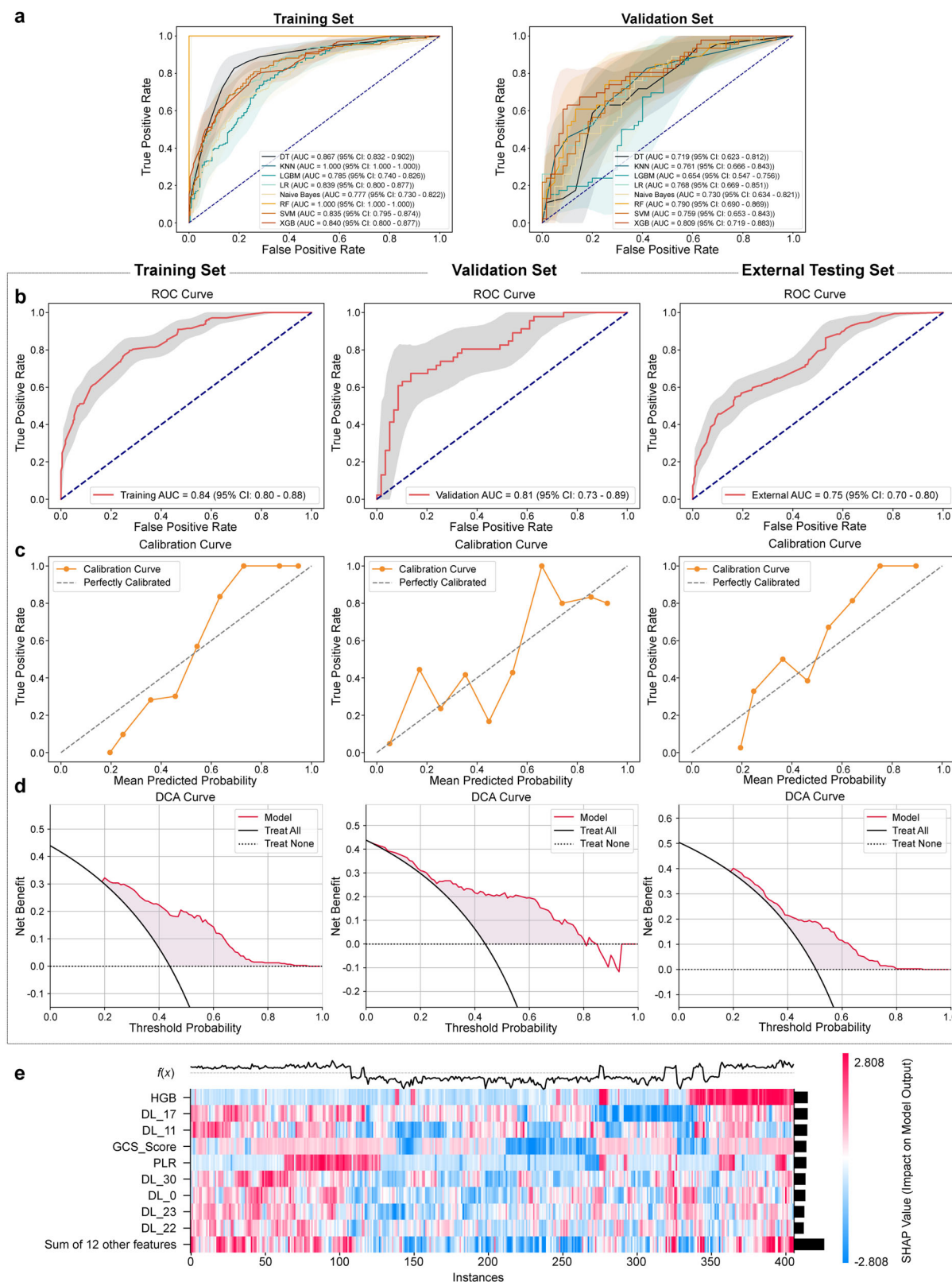


Fig. 9 | Performance evaluation of the simplified multiomics prediction model.

a ROC curves showing algorithm performance in the training and internal testing sets, including the DT, KNN, LGBM, LR, naive Bayes, RF, SVM, and XGB models, with corresponding AUC values and 95% confidence intervals. **b** ROC curves demonstrating model performance in the training, validation and merged external testing cohorts (AUC: 0.84, 0.81, and 0.75, respectively; gray shading indicates 95% CIs). **c** Calibration curves comparing the predicted probabilities with the observed

outcomes across the three cohorts (orange lines). **d** Decision curve analysis showing the net clinical benefits of the models (red lines) alongside the treat-all (solid black) and treat-none (dotted) strategies across different threshold probabilities. **e** SHAP value heatmap displaying the impact of key features (HGB, DL_17, DL_11, GCS_Score, PLR, DL_30, DL_0, DL_23) on model prediction; the corresponding model output distribution is shown in $f(x)$.

for blood bank reserves, although its clinical utility still requires further validation through prospective studies⁴⁴.

Model development based on clinical phenotypes represents the most common paradigm for machine learning in medicine. In our initial phase, we integrated clinical guidelines with frontline expertise to ensure appropriate feature selection (<https://www.facs.org/media/vgfgjpfk/best-practices-guidelines-traumatic-brain-injury.pdf>). However, most guideline criteria (such as coma from traumatic hematoma or posterior fossa lesions) are derived primarily from CT findings or clinical assessments, making them difficult to directly quantify from phenotypic data. Our final model, constructed using only three parameters—pupillary response, consciousness level, and age—achieved excellent robustness (external testing sets 1–3 AUC: 0.75). This finding is highly consistent with clinical understanding: pupillary response is closely related to the GCS score, while the level of consciousness serves as a simplified indicator for intracranial pressure. According to the literature, GCS^{45,46}, intracranial pressure^{47–49}, and age^{50–53} are all recognized determinants in neurosurgical decision-making. Therefore, we consider this streamlined model a further refinement of clinical expertise. For transfusion prediction, SHAP analysis revealed elevated lactate levels, low GCS scores, and decreased HGB levels as the three most significant predictors of transfusion requirements, which is consistent with the European guidelines on posttraumatic severe bleeding management⁵⁴ and the findings of multiple studies^{55–57}, which emphasize lactate as a sensitive marker for hemorrhage and tissue perfusion. GCS scores and HGB reflect the overall severity of TBI and anemic status, respectively, in accordance with our clinical experience. However, the model's low generalizability in predicting transfusion, despite feature selection from 146 admission variables, reflects the inherent challenge in predicting the future need for transfusion on the basis solely of early admission indicators (Merged external AUC: 0.69). These findings suggest that clinical phenotypic data alone may be insufficient for fully characterizing patients' transfusion needs.

The implementation of radiological imaging data through 2.5D deep transfer learning for predicting neurosurgical indicators is a relatively novel technique in brain imaging analysis. This approach offers a strategic balance between traditional methodologies; 2D approaches can capture only planar information, whereas 3D methods, although more comprehensive, are limited by computational challenges related to parameter complexity. Therefore, 2.5D methods leverage multiple consecutive slices to combine the advantages of both approaches^{58–60}. To address the inherent “black box” nature of deep learning, we employed Grad-CAM visualization, a technique that is being implemented more frequently for intracerebral hemorrhage prediction models⁶¹, to elucidate the decision-making processes of the models. Our results demonstrated significant spatial correspondence between MobileNet V2's feature extraction regions and areas of cerebral hemorrhage, suggesting that the 2.5D deep learning approach successfully captured trauma-related features across multiple planes of the brain. While the model achieved greater performance than the CBI model in the validation set, its lower stability across the external testing sets (AUC range: 0.70–0.81) highlights the challenges in maintaining consistent performance across diverse clinical settings.

The automation of clinical text extraction for predictive modeling is an innovative approach in current clinical prediction frameworks, potentially capturing physicians' comprehensive symptom descriptions, including severity assessments and multisystem injury patterns. Initially, we explored popular NLP models such as BERT for feature extraction or prediction output⁶². However, this approach was ultimately abandoned for two reasons: first, the admission diagnoses proved too concise for effective BERT implementation (10–50 characters in length), and second, more complex NLP models presented significant challenges in interpreting the results. Instead, we opted for a structured approach involving the conversion of admission diagnoses into structured data through stemming techniques. While this model demonstrated relatively modest predictive abilities in both surgical indication and transfusion prediction and suboptimal discrimination and calibration across the external datasets, it provided suitable transparency regarding the decision processes. Notably, these model features

identified through data-driven selection processes showed high concordance with previous clinical findings or experience, which are well supported by the existing medical literature. For example, brainstem injury and cerebral hemorrhage have emerged as the strongest surgical intervention indicators⁶³, whereas brainstem injury^{64,65}, pulmonary edema⁶⁶, and sacroiliac joint fracture have been identified as markers of poor disease prognosis or severe injury, potentially indirectly predicting the need for perioperative or post-operative transfusion. Future research with larger datasets and advances in reasoning-based transformer models may provide promising directions for enhancing clinical text analysis in TBI prediction tasks.

Previous multimodal studies in TBI research have primarily focused on prognostic applications, typically integrating imaging data with clinical phenotypic parameters but rarely incorporating clinical text features. For instance, recent work combined diffusion tensor imaging metrics with serum biomarkers for evaluating recovery outcomes in mild TBI patients, yet concentrated on long-term functional outcomes rather than acute clinical decision-making^{20,23,67}. Therefore, the study's most important result is the integration of clinical, radiological, and text data into the MDF model. By synthesizing complementary information from multiple data modalities, the fusion models consistently outperform the single-domain models. Specifically, radiological imaging data capture localized injury severity, clinical biomarkers reflect systemic physiological states, and medical text data provide a holistic summary of injury patterns and comorbidities. This comprehensive characterization allowed the models to achieve superior generalizability and robustness across diverse patient populations. SHAP analysis further elucidated the contributions of each data modality. For surgical prediction, radiological features are the most important, reflecting their critical role in assessing intracranial lesions. Conversely, transfusion prediction was driven primarily by clinical biomarkers, emphasizing their relevance in estimating systemic blood loss and hemodynamic compromise. These results indicate that while the multiomics model may not be optimal in all scenarios, it consistently demonstrates the highest robustness. Moreover, the model's end-to-end design requires no additional diagnostic tests for patients, enhancing its clinical feasibility. To further improve its applicability in emergency medical scenarios, we developed a simplified version of the multiomics model that omits blood biochemistry, coagulation function, and blood gas analysis results. Despite its reduced complexity, the simplified model maintained strong predictive performance, demonstrating its potential utility in resource-limited settings.

A key advantage of our multi-omics approach lies in its exceptional potential for seamless integration with existing emergency clinical workflows. Model prediction occurs after patients complete necessary admission examinations, typically within 2–4 h post-admission when all required features are available. Our results demonstrate that the model can generate outputs a median of 3 h before actual intervention, thereby providing additional preparation time for surgical implementation and massive transfusion protocols. When predictions indicate potential perioperative massive transfusion needs, multi-disciplinary teams receive alerts and blood banks initiate proactive resource allocation with products maintained in “standby” status. It is important to emphasize that actual transfusion decisions still depend on real-time physiological status during surgery, following current guidelines that recommend restrictive transfusion strategies for hemodynamically stable TBI patients (hemoglobin <7 g/dL)⁶⁸. The core value of our model lies in ensuring readily available blood products when healthcare providers determine massive transfusion is required. Furthermore, the model demonstrates remarkable flexibility when facing suboptimal data quality—for example, when CT imaging is compromised, it can operate in single-modal mode using only clinical biomarkers or text data while still providing clinically relevant results.

To facilitate the practical implementation of this model, we propose a three-layer system architecture integration framework comprising a data extraction layer, prediction engine, and clinical interface (Supplementary Note 1). This architecture addresses key implementation requirements through optimized design: (1) text information is extracted solely from patient admission diagnoses, significantly reducing data acquisition

complexity; (2) structured data employs standardized strategies for direct retrieval, minimizing processing burden; (3) image data processing utilizes the lightweight ShuffleNet v2 model, which supports smartphone operation. All input data derive from routine clinical activities and can be instantly accessed within existing electronic health record (EHR) systems without requiring additional examinations. The model features minimal computational load, with a user interface that can be seamlessly integrated into EHR platforms to provide real-time decision support without disrupting existing workflows. Currently, comprehensive software based on this model is under development, and we anticipate that this intelligent decision support system will advance to clinical prospective validation as soon as possible. To ensure effective application, we will provide structured training for users, covering core components including model principle understanding, output interpretation, limitation identification, and clinical knowledge integration (Supplementary Note 2). Supplementary Note 3 demonstrates the practical clinical application value of this model through four representative cases.

This study has several limitations that should be considered. First, the retrospective design introduces inherent limitations, including potential biases from missing data (deletion/imputation) and patient selection criteria, thereby affecting the representativeness of the study population. These limitations may influence machine learning model development by highlighting associations rather than causal relationships between features and predicted outcomes, ultimately impacting the models' generalizability and clinical applicability in diverse settings. Furthermore, the retrospective design limited our ability to capture comprehensive socioeconomic data, which may introduce unmeasured confounding bias. To mitigate these limitations, we implemented standardized review processes with independent physician assessment, mandatory data collection training, and unified treatment protocols across participating high-level tertiary institutions. Although these measures reduce, but may not eliminate, residual confounding from unmeasured socioeconomic factors. Regarding demographic representativeness, our analysis of age, sex, and other demographic factors across cohorts (Table 1) revealed relatively minor differences, with female proportions ranging from 21% to 32% and median ages between 51 and 57 years. To address potential demographic biases, our time-based inclusion strategy enrolled all eligible patients within predefined time windows, minimizing selection bias while representing the natural demographic distribution of TBI patients in real-world settings. Through our data-driven feature selection process, neither gender nor age emerged as input features in the final predictive models, suggesting a limited impact on model predictions. Additional subgroup analyses demonstrated consistent model performance across most demographic categories, although some subgroups could not undergo complete validation due to insufficient sample sizes (Supplementary Fig. 6). Our expanded diagnostic performance assessment across different age and sex subgroups strengthens evidence regarding model "fairness," complying with the TRIPOD-AI guidelines for model evaluation. Finally, despite robust retrospective multicenter validation, prospective real-time validation remains essential to confirm clinical utility and assess real-world performance under actual clinical⁴⁴.

In summary, this study demonstrates the feasibility and efficacy of a multimics fusion model that integrates clinical phenotypic, radiological imaging, and medical text data for TBI management. By synthesizing traditional clinical markers with deep learning features and medical text data, the model achieves an optimal balance among predictive accuracy, interpretability, and clinical applicability across diverse datasets. The automated integration and standardized decision-making processes render these models particularly valuable in emergency medical scenarios, such as major disasters and military conflicts, effectively facilitating medical resource optimization. Through prospective validation with larger and more diverse patient cohorts, these models demonstrate promise for broader clinical implementation, ultimately enhancing the quality of care for TBI patients.

Methods

Study design and population

We conducted a retrospective, multicenter study of 2219 TBI patients across four independent cohorts from January 2015 to June 2024. The study population comprised Cohort 1 ($n = 1060$) from the main campus of the Second Hospital of Hebei Medical University, Cohort 2 ($n = 501$) from the Seventh Medical Center of PLA General Hospital, Cohort 3 ($n = 262$) from the 960th Hospital of the PLA Joint Logistics Support Force, and Cohort 4 ($n = 396$) from the Luquan Campus of the Second Hospital of Hebei Medical University. All the participating centers are Grade IIIA hospitals with specialized neurosurgical units and emergency departments, and are selected to ensure diverse patient populations and healthcare settings for robust model development and validation.

Cohort 1 was designated for model development and internal validation, whereas Cohorts 2–4 were used for external testing (Supplementary Fig. 1). The inclusion criteria included a confirmed TBI diagnosis by qualified physicians and complete head CT scan data. Patients were excluded if they were <18 years, >80 years, or pregnant; transferred >48 h postinjury; received prior blood transfusion at other facilities; or had severe infectious diseases, active malignancies, chronic conditions, or other major illnesses affecting daily activities. All patients received standardized care according to current TBI management guidelines, with consistent surgical intervention criteria and blood transfusion protocols across all participating centers (detailed in Supplementary Note 4).

Data collection and outcomes

Data were collected from electronic health records across three domains: clinical phenotypic data, radiological imaging data, and medical text data. To ensure data consistency across participating centers, we implemented comprehensive standardization protocols: unified training documentation was developed for all data collectors with written specifications for inclusion/exclusion criteria and standardized data entry procedures. All data collection personnel underwent mandatory training sessions before study initiation. An independent quality control team performed systematic random audits of collected data to prevent entry errors and ensure consistent application of study criteria. Additionally, standardized review processes with mandatory dual physician assessment were implemented for all clinical decisions, with a third independent physician consulted for final adjudication in cases of disagreement. Variables with >30% missing data were excluded, with remaining missing values imputed using the missForest algorithm.

Two primary prediction targets were established: (1) Surgical indicators, determined through comprehensive clinical assessment; and (2) Blood transfusion requirement, defined as administration of >4 units of RBCs during surgery or within one week postoperatively. All medical imaging data and examination records were independently reviewed and validated by two designated clinical physicians. Predictor variable assessment was performed by neurosurgical clinicians with associate senior professional titles or above. The outcome assessors were blinded to patients' clinical outcomes during the evaluation process. In cases of disagreement, an independent third physician was consulted for final adjudication.

Model development overview

We developed two sets of machine learning models for predicting surgical indication and transfusion requirements. Each prediction target utilized four distinct models: Clinical Biomarker Integrative (CBI), Neural Radiological Imaging (NRI), Clinical Text Mining (CTM), and Multimics Data Fusion (MDF) models. A simplified transfusion prediction model was also developed for emergency settings. Eight ML algorithms were evaluated using Python's scikit-learn library (v3.8), including decision tree (DT), k-nearest neighbors (KNN), light gradient boosting machine (LGBM), logistic regression (LR), naive Bayes, random forest (RF), support vector machine (SVM), and eXtreme gradient boosting (XGB).

The study incorporated three main categories of predictor variables: clinical phenotypic data, radiological imaging data, and medical text data. Clinical phenotypic data comprised initial assessment parameters (pupillary response, level of consciousness, age, and sex) and 146 clinical variables spanning five categories: baseline characteristics, biochemical test results, coagulation profiles, blood gas analyses, and complete blood counts (Supplementary Table 1). The radiological data consisted of cranial CT scans, whereas the medical text data included admission diagnoses from clinical records. These variables were selected on the basis of their clinical relevance, immediate availability in emergency settings, and support from the literature (detailed in Supplementary Note 4).

Compliance with reporting guidelines

The development, validation, and reporting of all the models adhered strictly to the TRIPOD + AI guidelines⁴³. The detailed methodologies and additional analyses are provided in Supplementary Table 2.

Development and validation of a clinical biomarker integrative model (CBI model)

The variables were transformed into binary, ordinal, nominal, or discrete formats based on predefined criteria (Supplementary Table 1). Feature selection was performed with univariate logistic regression ($p < 0.05$), followed by multivariate validation. Eight machine learning algorithms were systematically evaluated using Python's scikit-learn library to determine optimal performance characteristics.

For model development, Cohort 1 data were stratified and randomly divided into training ($n = 848$, 80%) and validation ($n = 212$, 20%) sets. The training set underwent bootstrap resampling (1000 iterations) to assess model stability. Different testing strategies were implemented for the surgical indicator and transfusion requirement models, with the latter using merged data from Cohorts 2–4 due to smaller patient numbers (Supplementary Fig. 1).

Development and Validation of a Neural Radiological Imaging Model (NRI Model)

To minimize cross-center imaging variability, we implemented standardized preprocessing protocols across all participating centers. The CT images underwent uniform preprocessing procedures including standardized resampling ($1 \times 1 \times 1 \text{ mm}^3$), consistent window adjustment parameters (width=100 HU, Level=50 HU), and automated brain segmentation via PAIR software using identical algorithms across all centers. Our systematic slice extraction protocol involved automated identification of the maximum region of interest (ROI) cross-sectional area, followed by standardized multi-slice extraction at predetermined pixel intervals (-20, -10, -5, 0, 5, 10, 20 pixels) to ensure consistent 2.5D representation across different scanner types and acquisition protocols. A 2.5D deep transfer learning model was developed via two stages:

Stage 1-Initial training and refinement: Cohort 1 was divided into initial training ($n = 638$, 60%), refinement ($n = 210$, 20%), and validation ($n = 212$, 20%) sets. CT slices at seven symmetric positions (-20, -10, -5, 0, 5, 10, 20 pixels) were used as input channels. Multiple architectures were evaluated to determine optimal development.

Stage 2-Feature extraction, dimensionality reduction and secondary training: MobileNet V2 was employed as the feature extractor, with features reduced to 36 components via PCA and LASSO regression or RFE. These features were subsequently used for secondary training on the refinement set ($n = 210$) and validated on the validation set ($n = 212$). Notably, for the transfusion prediction model, features were re-extracted from the training set via the pretrained MobileNet V2 obtained from the surgical indicator prediction model, followed by direct training of machine learning models.

Development and Validation of the Clinical Text Mining Model (CTM Model)

The automated preprocessing pipeline included Chinese text segmentation (Jieba library), standardized lemmatization (KIMI system), and

transformation into structured binary features (Supplementary Table 3). The CTM model automatically extracted 44 distinct features from admission texts.

The Boruta algorithm was used for feature selection, followed by ML model construction following the CBI model methods.

Development and Validation of a Multiomics Data Fusion Model (MDF Model) and Simplified Version

The integrated model aggregated significant features from the CBI, NRI, and CTM models, with secondary feature selection via LASSO regression. Eight ML algorithms were evaluated following hyperparameter optimization. The surgical indicator model used only the refinement set ($n = 210$) from Cohort 1 to prevent data leakage (Supplementary Fig. 1). A simplified version incorporated only routine complete blood count parameters, CT imaging features, and admission text data, following the MDF model methodology.

Model Interpretability

SHAP quantified feature importance in surgical intervention and blood transfusion prediction models. Grad-CAM visualization identified relevant regions in radiological images. SHAP analysis was omitted for the LR models because of its inherent interpretability.

Statistical Analysis and Model Evaluation

Continuous variables are presented as medians (IQRs) or means $\pm$ SDs based on distribution, while categorical variables are expressed as numbers and percentages. Model performance was evaluated using multiple metrics including AUC-ROC, sensitivity, specificity, and F1 score, with 95% CIs calculated via bootstrapping (1000 resamples). Calibration was assessed using calibration plots and the Hosmer-Lemeshow test, while clinical utility was evaluated through decision curve analysis. Correlations between predicted and actual transfusion volumes were assessed using Pearson's correlation coefficient. P values were adjusted for multiple testing using the Benjamini-Hochberg method, and model comparisons were performed using DeLong's test.

The sample size calculation followed the events per variable principle, requiring a minimum of 10 events per predictor variable. With 15 candidate predictors and an anticipated 50% event rate, the minimum required sample size was 300 patients. Our development cohort ($n = 848$) substantially exceeded this requirement, ensuring adequate statistical power for both development and validation.

Ethics statement

This study received institutional review board approval from all participating centers. The Research Ethics Committee of the Second Hospital of Hebei Medical University approved data collection for Cohorts 1 and 4 (2023-R502). Data collection for Cohort 2 was approved by the Research Ethics Committee of the Seventh Medical Center of PLA General Hospital (S2023-023-01), while Cohort 3 was approved by the Research Ethics Committee of the 960th Hospital of the PLA Joint Logistics Support Force (2023-073). All patient data underwent anonymization and deidentification prior to analysis. The requirement for informed consent was waived due to the study's retrospective nature. All procedures adhered to the Declaration of Helsinki and applicable ethical guidelines for medical research.

Data availability

Two MDF models are available: one for predicting surgical indications (<https://www.xsmartanalysis.com/model/views?id=171>) and the other for predicting transfusion needs (<https://www.xsmartanalysis.com/model/views?id=172>). These web-based tools allow users to validate our models or analyze their own data following standardized protocols. The platform ensures data security through Alibaba Cloud servers and dual backup mechanisms, with encrypted storage for all uploaded data. The multimodal models that include clinical phenotypic data, radiological imaging findings, and medical text data from multiple centers used in this study are not publicly available due to patient privacy protection policies. However,

researchers interested in accessing the data for scientific research purposes may submit requests to the corresponding author. These requests will be reviewed within 3 months and subject to approval by the Ethics Committees of three participating hospitals: the Second Hospital of Hebei Medical University, the Seventh Medical Center of PLA General Hospital, and the 960th Hospital of the PLA Joint Logistics Support Force. The Ethics Committees reserve the right to deny access to the data.

Code availability

The complete source code for model construction and implementation, including data preprocessing pipelines, model architecture, and evaluation scripts, is available for download through Baidu Netdisk (<https://pan.baidu.com/s/1bRR3hD4WgwC3IXgr8o8zLw>) via the access code jxwk. This code repository contains detailed documentation and usage instructions to facilitate reproduction of our results and further development of the models.

Received: 3 March 2025; Accepted: 7 October 2025;

Published online: 19 November 2025

References

- GBD 2016 Traumatic Brain Injury and Spinal Cord Injury Collaborators. Global, regional, and national burden of traumatic brain injury and spinal cord injury, 1990–2016: a systematic analysis for the Global Burden of Disease Study 2016. *Lancet Neurol.* **18**, 56–87 (2019).
- Dewan, M. C. et al. Estimating the global incidence of traumatic brain injury. *J. Neurosurg.* **130**, 1080–1097 (2019).
- Helmick, K. M. et al. Traumatic brain injury in the US military: epidemiology and key clinical and research programs. *Brain Imaging Behav.* **9**, 358–366 (2015).
- Wang, K., Cui, D. & Gao, L. Traumatic brain injury: a review of characteristics, molecular basis and management. *Front. Biosci. (Landmark Ed.)* **21**, 890–899 (2016).
- Ma, Z. et al. Traumatic brain injury in elderly population: a global systematic review and meta-analysis of in-hospital mortality and risk factors among 2.22 million individuals. *Ageing Res. Rev.* **99**, 102376 (2024).
- Forslund, M. V. et al. Health-related quality of life trajectories across 10 years after moderate to severe traumatic brain injury in Norway. *J. Clin. Med.* **10**, 157 (2021).
- Freire, M. et al. Cellular and molecular pathophysiology of traumatic brain injury: what have we learned so far. *Biology (Basel)* **12**, 1139 (2023).
- Stocchetti, N. & Zanier, E. R. Chronic impact of traumatic brain injury on outcome and quality of life: a narrative review. *Crit. Care (Lond., Engl.)* **20**, 148 (2016).
- AlSowaiegh, R. et al. The Emergency Surgery Score is a powerful predictor of outcomes across multiple surgical specialties: results of a retrospective nationwide analysis. *Surgery* **170**, 1501–1507 (2021).
- Buccilli, B. et al. Neuroprotection: surgical approaches in traumatic brain injury. *Surg. Neurol. Int.* **15**, 23 (2024).
- Rakhit, S. et al. Management and challenges of severe traumatic brain injury. *Semin. Respir. Crit. Care Med.* **42**, 127–144 (2021).
- Turgeon, A. F. et al. Liberal or restrictive transfusion strategy in patients with traumatic brain injury. *N. Engl. J. Med.* **391**, 722–735 (2024).
- Taccone, F. S. et al. Restrictive vs liberal transfusion strategy in patients with acute brain injury: the TRAIN Randomized Clinical Trial. *J. Am. Med. Assoc.* **332**, 1623–1633 (2024).
- Lee, S. M. et al. Development and validation of a prediction model for need for massive transfusion during surgery using intraoperative hemodynamic monitoring data. *JAMA Netw. Open* **5**, e2246637 (2022).
- Eloranta, S. & Boman, M. Predictive models for clinical decision making: Deep dives in practical machine learning. *J. Intern. Med.* **292**, 278–295 (2022).
- Gauss, T. et al. Pilot deployment of a machine-learning enhanced prediction of need for hemorrhage resuscitation after trauma - the ShockMatrix pilot study. *BMC Med. Inform. Decis. Mak.* **24**, 315 (2024).
- Hadijski, L. et al. AAPM task group report 273: recommendations on best practices for AI and machine learning for computer-aided diagnosis in medical imaging. *Med. Phys.* **50**, e1–e24 (2023).
- Yin, A. A. et al. Machine learning models for predicting in-hospital outcomes after non-surgical treatment among patients with moderate-to-severe traumatic brain injury. *J. Clin. Neurosci.* **120**, 36–41 (2024).
- Monteiro, M. et al. Multiclass semantic segmentation and quantification of traumatic brain injury lesions on head CT using deep learning: an algorithm development and multicentre validation study. *Lancet Digital Health* **2**, e314–e322 (2020).
- Pease, M. et al. Outcome prediction in patients with severe traumatic brain injury using deep learning from head CT scans. *Radiology* **304**, 385–394 (2022).
- Czeiter, E. et al. Blood biomarkers on admission in acute traumatic brain injury: Relations to severity, CT findings and care path in the CENTER-TBI study. *EBioMedicine* **56**, 102785 (2020).
- Richter, S. et al. Prognostic value of serum biomarkers in patients with moderate-severe traumatic brain injury, differentiated by marshall computer tomography classification. *J. Neurotrauma* **40**, 2297–2310 (2023).
- Richter, S. et al. Predicting recovery in patients with mild traumatic brain injury and a normal CT using serum biomarkers and diffusion tensor imaging (CENTER-TBI): an observational cohort study. *EClinicalMedicine* **75**, 102751 (2024).
- Siqueira Pinto, M. et al. Use of support vector machines approach via ComBat harmonized diffusion tensor imaging for the diagnosis and prognosis of mild traumatic brain injury: a CENTER-TBI study. *J. Neurotrauma* **40**, 1317–1338 (2023).
- Chu, J., Dong, W., Wang, J., He, K. & Huang, Z. Treatment effect prediction with adversarial deep learning using electronic health records. *BMC Med. Inform. Decis. Mak.* **20**, 139 (2020).
- Pham, T., Tran, T., Phung, D. & Venkatesh, S. Predicting healthcare trajectories from medical records: a deep learning approach. *J. Biomed. Inform.* **69**, 218–229 (2017).
- Rajkomar, A. et al. Scalable and accurate deep learning with electronic health records. *NPJ Digit. Med.* **1**, 18 (2018).
- Habibzadeh, A. et al. Machine learning-based models to predict the need for neurosurgical intervention after moderate traumatic brain injury. *Health Sci. Rep.* **6**, e1666 (2023).
- Moyer, J. D. et al. Machine learning-based prediction of emergency neurosurgery within 24 h after moderate to severe traumatic brain injury. *World J. Emerg. Surg.* **17**, 42 (2022).
- Gelderblom, M. E., Stevens, K., Houterman, S., Weyers, S. & Schoot, B. C. Prediction models in gynaecology: transparent reporting needed for clinical application. *Eur. J. Obstet. Gynecol. Reprod. Biol.* **265**, 190–202 (2021).
- Groot, O. Q. et al. Availability and reporting quality of external validations of machine-learning prediction models with orthopedic surgical outcomes: a systematic review. *Acta Orthop.* **92**, 385–393 (2021).
- Papadomanolakis-Pakis, N. et al. Prognostic clinical prediction models for acute post-surgical pain in adults: a systematic review. *Anaesthesia* **79**, 1335–1347 (2024).
- Warman, A., Kalluri, A. L. & Azad, T. D. Machine learning predictive models in neurosurgery: an appraisal based on the TRIPOD guidelines. Systematic review. *Neurosurg. Focus* **54**, E8 (2023).
- Yang, L. et al. Reporting of coronavirus disease 2019 prognostic models: the transparent reporting of a multivariable prediction model for individual prognosis or diagnosis statement. *Ann. Transl. Med.* **9**, 421 (2021).
- Boehm, K. M., Khosravi, P., Vanguri, R., Gao, J. & Shah, S. P. Harnessing multimodal data integration to advance precision oncology. *Nat. Rev. Cancer* **22**, 114–126 (2022).

36. Lobato-Delgado, B., Priego-Torres, B. & Sanchez-Morillo, D. Combining molecular, imaging, and clinical data analysis for predicting cancer prognosis. *Cancers (Basel)* **14**, 3215 (2022).
37. Zhou, K. et al. Integration of multimodal data from disparate sources for identifying disease subtypes. *Biology (Basel)* **11**, 360 (2022).
38. Petch, J., Di, S. & Nelson, W. Opening the Black Box: the promise and limitations of explainable machine learning in cardiology. *Can. J. Cardiol.* **38**, 204–213 (2022).
39. Rudin, C. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nat. Mach. Intell.* **1**, 206–215 (2019).
40. Cole, E. et al. A decade of damage control resuscitation: new transfusion practice, new survivors, new directions. *Ann. Surg.* **273**, 1215–1220 (2021).
41. Dorken-Gallastegi, A. et al. Whole blood and blood component resuscitation in trauma: interaction and association with mortality. *Ann. Surg.* **280**, 1014–1020 (2024).
42. Gauss, T. et al. Comparison of machine learning and human prediction to identify trauma patients in need of hemorrhage control resuscitation (ShockMatrix study): a prospective observational study. *Lancet Reg. Health Eur.* **55**, 101340 (2025).
43. Collins, G. S. et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *Br. Med. J. (Clin. Res. Ed.)* **385**, e078378 (2024).
44. Berkhout, W. et al. Operationalization of Artificial Intelligence Applications in the Intensive Care Unit: a systematic review. *JAMA Netw. Open* **8**, e2522866 (2025).
45. McNett, M. A review of the predictive ability of Glasgow Coma Scale scores in head-injured patients. *J. Neurosci. Nurs.* **39**, 68–75 (2007).
46. Tude Melo, J. R. et al. Criteria for neurosurgical treatment of children and adolescents with traumatic brain injury in a Brazilian level 1 trauma center. *J. Neurosurg. Pediatr.* **35**, 72–78 (2025).
47. Kalyvas, A. et al. A systematic review of surgical treatments of idiopathic intracranial hypertension (IIH). *Neurosurg. Rev.* **44**, 773–792 (2021).
48. Sahuquillo, J. & Dennis, J. A. Decompressive craniectomy for the treatment of high intracranial pressure in closed traumatic brain injury. *Cochrane Database Syst. Rev.* **12**, CD003983 (2019).
49. Zhang, D., Sheng, Y., Wang, C., Chen, W. & Shi, X. Global traumatic brain injury intracranial pressure: from monitoring to surgical decision. *Front. Neurol.* **15**, 1423329 (2024).
50. Barthélemy, E. J., Melis, M., Gordon, E., Ullman, J. S. & Germano, I. M. Decompressive craniectomy for severe traumatic brain injury: a systematic review. *World Neurosurg.* **88**, 411–420 (2016).
51. Lee, K. S. et al. Surgical decision making for the elderly patients in severe head injuries. *J. Korean Neurosurg. Soc.* **55**, 195–199 (2014).
52. Pilitsis, J. et al. Outcomes in octogenarians with subdural hematomas. *Clin. Neurol. Neurosurg.* **115**, 1429–1432 (2013).
53. Unterhofer, C., Ho, W. M., Wittlinger, K., Thomé, C. & Ortler, M. “I am not afraid of death” — a survey on preferences concerning neurosurgical interventions among patients over 75 years. *Acta Neurochir. (Wien.)* **159**, 1547–1552 (2017).
54. Rossaint, R. et al. The European guideline on management of major bleeding and coagulopathy following trauma: sixth edition. *Crit. Care (Lond., Engl.)* **27**, 80 (2023).
55. Fu, Y. Q., Bai, K. & Liu, C. J. The impact of admission serum lactate on children with moderate to severe traumatic brain injury. *PLoS ONE* **14**, e0222591 (2019).
56. Lekomtseva, Y. Targeting higher levels of lactate in the post-injury period following traumatic brain injury. *Clin. Neurol. Neurosurg.* **196**, 106050 (2020).
57. Wang, R., He, M., Qu, F., Zhang, J. & Xu, J. Lactate albumin ratio is associated with mortality in patients with moderate to severe traumatic brain injury. *Front. Neurol.* **13**, 662385 (2022).
58. Li, M. et al. Enhancing automatic prediction of clinically significant prostate cancer with deep transfer learning 2.5-dimensional segmentation on bi-parametric magnetic resonance imaging (bp-MRI). *Quant. Imaging Med. Surg.* **14**, 4893–4902 (2024).
59. Ottesen, J. A. et al. 2.5D and 3D segmentation of brain metastases with deep learning on multinational MRI data. *Front. Neuroinform.* **16**, 1056068 (2022).
60. Zeng, Y. et al. A 2.5D deep learning-based method for drowning diagnosis using post-mortem computed tomography. *IEEE J. Biomed. Health Inf.* **27**, 1026–1035 (2023).
61. Yeo, M. et al. Evaluation of techniques to improve a deep learning algorithm for the automatic detection of intracranial haemorrhage on CT head imaging. *Eur. Radiol. Exp.* **7**, 17 (2023).
62. Liu, H. et al. Use of BERT (Bidirectional Encoder Representations from Transformers)-Based Deep Learning Method for Extracting Evidences in Chinese Radiology Reports: development of a computer-aided liver cancer diagnosis framework. *J. Med. Internet Res.* **23**, e19689 (2021).
63. Vakil, M. T. & Singh, A. K. A review of penetrating brain trauma: epidemiology, pathophysiology, imaging assessment, complications, and treatment. *Emerg. Radiol.* **24**, 301–309 (2017).
64. O’Phelan, K. H. Not Always a Nail in the Coffin! Brainstem lesions after traumatic brain injury. *Neurocrit. Care* **35**, 306–307 (2021).
65. Williams, J. R. et al. Prognostic value of hemorrhagic brainstem injury on early computed tomography: a TRACK-TBI study. *Neurocrit. Care* **35**, 335–346 (2021).
66. Maslonka, M. A., Sheehan, K. N., Datar, S. V., Vachharajani, V. & Namen, A. Pathophysiology and management of neurogenic pulmonary edema in patients with acute severe brain injury. *South. Med. J.* **115**, 784–789 (2022).
67. Khalili, H. et al. Prognosis prediction in traumatic brain injury patients using machine learning algorithms. *Sci. Rep.* **13**, 960 (2023).
68. Carson, J. L. et al. Red blood cell transfusion: 2023 AABB International Guidelines. *J. Am. Med. Assoc.* **330**, 1892–1902 (2023).

Acknowledgements

This work was supported by the National Natural Science Foundation of China Youth Project (82200738).

Author contributions

J.D., T.D., Y.Z. and N.Z. conceptualized the study; performed data curation, formal analysis, investigation, and methodology development; and wrote the original draft. K.L., C.W., Z.Z. and J.M. contributed to software development, data validation, and visualization. Ha.W., L.L., and P.M. provided supervision, project administration, and critical revision of the manuscript. Y.Z., T.W., X.H., and J.D. verified the underlying data and were responsible for the overall direction of the project. All the authors read and approved the final version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41746-025-02072-5>.

Correspondence and requests for materials should be addressed to Jiang Deng, Xiang-Yan Huang, Tao Wu or Yan-Yu Zhang.

Reprints and permissions information is available at <http://www.nature.com/reprints>

Publisher’s note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025